

MMEC OFFICIAL SYLLABUS 2025-26

Basic Electrical & Electronics Engineering

Complete 4-Unit Lecture Notes & University Solved Examination Bank

Course Code: BELE-001 / BEEE

Department: B.Tech (First Year Common Curriculum)

Units Covered: Units 1, 2, 3, and 4 (Exhaustive Theory + Numerical Problems)

Mathematical Rigor: Publication-grade KaTeX typesetting

Verified: Step-by-Step Numerical Calculations & Marking Scheme Included

Generated for Maharishi Markandeshwar Engineering College (MMEC) Students • 100% Offline Accessible

DC Circuits & AC Circuit Analysis (Made Super Simple)

Exam Weightage: 24 - 28 Marks • **Estimated Study Time:** 35 mins

Circuit analysis explained with water pipes! Kirchhoff's Laws (Charge & Energy conservation), Thevenin's magic shortcut, Maximum Power bicycle analogy, and AC waves with R-L-C resonance.

1. Voltage, Current & Electrical Sources (The Water Pipe Model)

1. Fundamental Principles of Circuit Theory vs Field Theory

Electrical engineering operates in two complementary realms: Maxwell's Electromagnetic Field Theory and Lumped Circuit Theory.

● **The Lumped Circuit Assumption:** Electric circuit theory is an idealized approximation of Maxwell's equations, valid when the physical dimension of the network d is much smaller than the electromagnetic wavelength $\lambda = c/f$:

$$d \ll \lambda = \frac{c}{f}$$

At power frequencies (50, Hz), $\lambda = \frac{3 \times 10^8}{50} = 6000$, km. Since bench circuits and domestic appliances are meters long, electromagnetic propagation delays are negligible.

Charge, Current & Energy Relations:

Electric Current ($i(t)$): The time rate of transport of electric charge past a specified cross-section:

$$i(t) = \frac{dq(t)}{dt} \Leftrightarrow q(t) = \int_{-\infty}^t i(\tau) d\tau [\text{Coulombs, C}]$$

Electric Potential / Voltage ($v(t)$): The work required to move a unit positive test charge through an electric field between two points:

$$v(t) = \frac{dw(t)}{dq} [\text{Joules/Coulomb} = \text{Volts, V}]$$

Instantaneous Electric Power ($p(t)$): By the differential chain rule:

$$p(t) = \frac{dw}{dt} = \left(\frac{dw}{dq} \right) \left(\frac{dq}{dt} \right) = v(t) \cdot i(t) [\text{Watts, W}]$$

- **Passive Sign Convention:** If reference current enters the positive terminal (+), $p(t) > 0 \Rightarrow$ **element absorbs / consumes energy**. If current leaves positive terminal, $p(t) < 0 \Rightarrow$ **element delivers / generates energy**.

2. Comprehensive Taxonomy of Electrical Energy Sources

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[Electrical Sources]

|

+-----+-----+

||

[Independent Sources] [Dependent (Controlled) Sources]

Terminal output fixed, Controlled by voltage or current

independent of circuit elsewhere in network

||

+-----+-----+ +-----+-----+

|||||

[V-Source] [I-Source] VCVS VCCS CCVS CCCS

...

Practical (Real-World) vs Ideal Sources:

Practical Voltage Source: Modeled as an ideal independent EMF V_s in series with an internal series resistance R_{se} :

$$V_L = V_s - I_L \cdot R_{se}$$

- As load current $I_L \rightarrow \infty$, terminal voltage droops linearly.
- For an **ideal voltage source**, internal resistance is zero: $R_{se} = 0 \Rightarrow V_L = V_s$ unconditionally.

Practical Current Source: Modeled as an ideal constant current generator I_s in parallel with an internal shunt resistance R_{sh} :

$$I_L = I_s - \frac{V_L}{R_{sh}}$$

- For an **ideal current source**, internal shunt resistance is infinite: $R_{sh} = \infty \Rightarrow I_L = I_s$ unconditionally.

The Source Transformation Equivalence Theorem:

A practical voltage source (V_s, R_{se}) is electrically identical at external terminals to a practical current source (I_s, R_{sh}) if and only if:

$$I_s = \frac{V_s}{R_{se}} \text{ and } R_{sh} = R_{se}$$

Problem:

A simple circuit has a 12, V battery connected across three series resistors: $R_1 = 2, \Omega$, $R_2 = 3, \Omega$, and $R_3 = 1, \Omega$. Find the circuit current I and voltage drop across each resistor.

GIVEN SCHEMATIC: 12V BATTERY CONNECTED TO 3 SERIES RESISTORS

+

–

$$V = 12 \text{ V}$$

$$I = ? \text{ A}$$

$$R_1 = 2.0 \, \Omega$$

$$R_2 = 3.0 \, \Omega$$

$$R_3 = 1.0 \, \Omega$$

Given: $V = 12 \text{ V}$, $R_1 = 2 \, \Omega$, $R_2 = 3 \, \Omega$, $R_3 = 1 \, \Omega \Rightarrow$ Find Total Current I and Drops V_1 , V_2 , V_3

Fig 1.1a: Given Series Resistor Circuit Network with 12V DC Source

Step 1 🚀 : Total Equivalent Resistance

$$R_{eq} = R_1 + R_2 + R_3 = 2 + 3 + 1 = \mathbf{6.0, \Omega}$$

Step 2 ⚡ : Circuit Current via Ohm's Law

$$I = \frac{V}{R_{eq}} = \frac{12, \text{V}}{6, \Omega} = \mathbf{2.0, \text{Amperes}}$$

Step 3 🎯 : Voltage Drops

$$V_1 = IR_1 = 2 \times 2 = \mathbf{4.0, \text{V}}, V_2 = IR_2 = 2 \times 3 = \mathbf{6.0, \text{V}}, V_3 = IR_3 = 2 \times 1 = \mathbf{2.0, \text{V}}$$

Sum verification: $4 + 6 + 2 = 12, \text{V}$.

SOLVED SCHEMATIC: EQUIVALENT LOOP CURRENT & INDIVIDUAL VOLTAGE DROPS

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$$V = 12\text{ V}$$

$$I = 2.0 \text{ A}$$

$$R_1 = 2.0 \, \Omega$$

$$V_1 = 4.0 \text{ V}$$

$$R_2 = 3.0 \, \Omega$$

$$V_2 = 6.0 \text{ V}$$

$$R_3 = 1.0 \, \Omega$$

$$V_3 = 2.0 \text{ V}$$

$$R_{\text{eq}} = 2 + 3 + 1 = 6.0 \, \Omega$$

$$\text{KVL Check: } \Sigma V_{\text{drops}} = 4\text{V} + 6\text{V} + 2\text{V} = 12\text{V} = V_{\text{supply}} \checkmark$$

Solved: Total Current $I = 2.0 \text{ A}$ | $V_1 = 4.0 \text{ V}$, $V_2 = 6.0 \text{ V}$, $V_3 = 2.0 \text{ V}$ (Energy Conserved)

Fig 1.1b: Solved Series Loop Showing Ohm's Law Voltage Drops and Energy Balance

🌀 Total Current $I = 2.0, \text{ A}$ | Drops: $V_1 = 4.0, \text{ V}$, $V_2 = 6.0, \text{ V}$, $V_3 = 2.0, \text{ V}$.

University Question (8 Marks):

A real battery with open-circuit EMF $E = 24, \text{ V}$ is connected to a load resistor $R_L = 9, \Omega$. The terminal voltage drops to $18, \text{ V}$. Determine: (1) Internal resistance R_{se} , (2) Current delivered, (3) Power delivered to the load.

GIVEN SCHEMATIC: PRACTICAL BATTERY DRIVING EXTERNAL LOAD RESISTOR

Practical Battery Model (E, R_{se})

+

—

$E = 24 \text{ V}$

$R_{se} = ? \Omega$

(Internal Resistance)

Terminal A (+)

Terminal B (−)

$$R_L = 9.0 \, \Omega$$

$$V_L = 18 \, \text{V}$$

Given: $E = 24 \, \text{V}$, $R_L = 9 \, \Omega$, $V_L = 18 \, \text{V} \Rightarrow$ Find Internal Resistance R_{se} , Current I_L , and Load Power P_L

Fig 1.1c: Practical Real Battery Circuit Showing Internal Resistance and Terminal Voltage Droop

Step 1 🚀 : Find the Load Current I_L

$$I_L = \frac{V_L}{R_L} = \frac{18, \text{V}}{9, \Omega} = \mathbf{2.0, A}$$

Step 2 ⚡ : Calculate Internal Resistance R_{se}

Internal voltage lost across electrolyte:

$$V_{\text{lost}} = E - V_L = 24 - 18 = 6.0, \text{V}$$

$$R_{se} = \frac{V_{\text{lost}}}{I_L} = \frac{6.0, \text{ V}}{2.0, \text{ A}} = \mathbf{3.0, \Omega}$$

Step 3 🎯 : Useful Power Delivered

$$P_L = I_L^2 \cdot R_L = (2.0)^2 \times 9 = \mathbf{36.0, \text{ Watts}}$$

SOLVED SCHEMATIC: COMPLETE BATTERY MODEL WITH VOLTAGE & POWER PARTITION

Solved Battery Internal Network

+

—

E = 24 V

$$R_{se} = 3.0 \, \Omega$$

$$V_{lost} = 6.0 \, V$$

$$I_L = 2.0 \, A$$

A (+)

B (-)

$$R_L = 9.0 \, \Omega$$

$$P_L = 36.0 \, W \, (\text{Useful})$$

$$V_L = 18 \, V$$

Solved: $R_{se} = 3.0 \, \Omega$ | Current $I_L = 2.0 \, A$ | $P_L = 36.0 \, W$ | Internal Loss = $12.0 \, W$ (Total $48 \, W$)

Fig 1.1d: Solved Equivalent Battery Model Confirming Internal Resistance & Load Power Balance

🔗 Current $I = 2.0, \text{ A}$ | Internal Resistance $R_{se} = 3.0, \Omega$ | Load Power $P_L = 36.0, \text{ W}$.

University Question (10 Marks):

A practical voltage source of $40, \text{ V}$ in series with $5, \Omega$ is connected in parallel with a practical current source of $6, \text{ A}$ in parallel with $10, \Omega$. The combined network feeds a load resistor $R_L = 10, \Omega$. Use source transformation to find the current I_L and power in R_L .

GIVEN SCHEMATIC: DUAL-SOURCE ACTIVE NETWORK DRIVING LOAD RESISTOR

Practical V-Source

+

–

40 V

$$R_1 = 5 \, \Omega$$

Practical I-Source

$$I_s = 6 \, \text{A}$$

$$R_2 = 10 \, \Omega$$

A

B

$$R_L = 10 \, \Omega$$

$$I_L = ?$$

University Challenge: Apply Source Transformation to determine Load Current I_L and Load Power P_L in R_L

Fig 1.1e: Multi-Source Active Network Requiring Source Transformation to Resolve Load Parameters

Step 1 🚀 : Convert Practical Voltage Source to Current Source

Ideal current: $I_{s1} = \frac{V_s}{R_{se}} = \frac{40, \text{V}}{5, \Omega} = 8.0, \text{A}$ (pointing up).

Parallel resistance: $R_{sh1} = 5.0, \Omega$.

Step 2 ⚡ : Combine Parallel Current Sources and Parallel Resistors

Net current source: $I_{\text{total}} = 8.0, \text{A} + 6.0, \text{A} = \mathbf{14.0, \text{A}}$.

Combined parallel resistance:

$$R_p = 5, \Omega \parallel 10, \Omega = \frac{5 \times 10}{5 + 10} = \frac{50}{15} = \mathbf{3.33, \Omega}$$

Step 3 🎯 : Transform Back to Single Thevenin Voltage Source

$V_{th} = I_{\text{total}} \times R_p = 14.0 \times \frac{10}{3} = \mathbf{46.67, \text{V}}$.

$R_{th} = R_p = 3.33, \Omega$.

Step 4 🎯 : Connect Load $R_L = 10, \Omega$ and Calculate I_L and P_L

$$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{46.67}{3.33 + 10} = \frac{46.67}{13.33} = \mathbf{3.50, \text{Amperes}}$$

$$P_L = I_L^2 \cdot R_L = (3.50)^2 \times 10 = \mathbf{122.5, \text{Watts}}$$

SOLVED SCHEMATIC: STEP-BY-STEP SOURCE TRANSFORMATION & THEVENIN EQUIVALENT

Stage 1: Parallel Current Combine

- $40\text{V} / 5\Omega \Rightarrow 8.0\text{ A} \parallel 5.0\Omega$

- In parallel with $6.0\text{ A} \parallel 10.0\Omega$

$$\Rightarrow I_{\text{total}} = 8\text{ A} + 6\text{ A} = 14.0\text{ A}$$

$$\Rightarrow R_{\text{p}} = 5 \parallel 10 = 3.33\Omega$$

- Reconvert to Series Thevenin:

$$V_{\text{th}} = 14\text{ A} \times 3.33\Omega = 46.67\text{ V}$$

Stage 2: Thevenin Equivalent + Load R_{L}

+

–

46.67 V

$$R_{th}=3.33\Omega$$

A

B

$$R_L=10\Omega$$

$$I_L=3.5A$$

Transformation Verified: $I_L = 46.67 / (3.33 + 10) = 3.50 \text{ A}$ | $P_L = (3.50)^2 \times 10 = 122.5 \text{ Watts}$

Fig 1.1f: Transformation Reduction Pipeline Yielding Thevenin Equivalent and Solved Load Power



Source Transformation Result: Load Current $I_L = 3.50, \text{ A}$ | Power $P_L = 122.5, \text{ W}$.

2. Kirchhoff's Laws (KCL & KVL Made Intuitive)

1. In-Depth Theoretical Foundations of Kirchhoff's Laws

German physicist Gustav Kirchhoff established the fundamental topological constraints governing all lumped electrical circuits.

A. Kirchhoff's Current Law (KCL) — Conservation of Electric Charge:

Formal Statement: The algebraic sum of currents entering any closed boundary or electrical node is identically zero at every instant of time:

$$\sum_{k=1}^N \mathbf{i}_k(\mathbf{t}) = \mathbf{0} \Leftrightarrow \sum \mathbf{i}_{\text{entering}} = \sum \mathbf{i}_{\text{leaving}}$$

Physical Derivation: By Maxwell's continuity equation $\nabla \cdot \mathbf{J} + \frac{\partial \rho_v}{\partial t} = 0$. Integrating over a closed Gaussian surface enclosing the node:

$$\oint_S \mathbf{J} \cdot d\mathbf{A} = -\frac{d}{dt} \int_V \rho_v dV = -\frac{dq_{\text{enclosed}}}{dt}$$

Since an electrical node has zero physical volume ($V \rightarrow 0$), it cannot store or accumulate net charge: $\frac{dq}{dt} = 0 \Rightarrow \sum i_k = 0$.

B. Kirchhoff's Voltage Law (KVL) — Conservation of Energy:

● **Formal Statement:** The algebraic sum of all branch voltages around any closed loop or mesh in an electric network is identically zero:

$$\sum_{k=1}^M \mathbf{v}_k(\mathbf{t}) = \mathbf{0} \Leftrightarrow \sum \mathbf{V}_{\text{rises}} = \sum \mathbf{V}_{\text{drops}}$$

Physical Derivation: In static and quasi-static electromagnetic fields, Faraday's Law states $\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt}$. In lumped circuits, magnetic flux linking the loop is confined inside discrete magnetic elements (inductors), so outside the elements $\frac{d\Phi_B}{dt} = 0 \Rightarrow \oint_C \mathbf{E} \cdot d\mathbf{l} = 0$, confirming potential is a conservative field.

2. Network Topology & Systematic Circuit Formulation

Let a planar network contain b branches and n nodes:

Independent KCL Node Equations: Exactly $n - 1$ independent equations (one node chosen as reference / ground).

Independent KVL Mesh Equations: By Euler's topological theorem:

$l = b - n + 1$ independent fundamental loops (meshes)

Matrix Formulation of Mesh Analysis:

$$[R][I] = [V]$$

$$\begin{bmatrix} R_{11} & R_{12} & \dots & R_{1m} \\ R_{21} & R_{22} & \dots & R_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ R_{m1} & R_{m2} & \dots & R_{mm} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_m \end{bmatrix} = \begin{bmatrix} V_{s1} \\ V_{s2} \\ \vdots \\ V_{sm} \end{bmatrix}$$

Self-resistance R_{ii} : Sum of all resistances in mesh i (always positive).

Mutual resistance R_{ij} ($i \neq j$): Negative sum of resistances shared between mesh i and mesh j (symmetric: $R_{ij} = R_{ji}$).

Driving voltage V_{si} : Net algebraic sum of voltage rises clockwise around mesh i .

Problem:

At an electrical junction, four wires meet. Currents entering the junction are $I_1 = 4$, A and $I_2 = 7$, A. One current leaving is $I_3 = 6$, A. Find the unknown current I_4 .

SCHEMATIC 1: SINGLE-NODE KCL CURRENT CONSERVATION

Gaussian Closed Boundary S (Zero Charge Storage)

$I_1 = 4.0 \text{ A}$ (Entering)

$I_2 = 7.0 \text{ A}$ (Entering)

Node A

$I_3 = 6.0 \text{ A}$ (Leaving)

$I_4 = ? \text{ A}$ (Leaving)

$$\Sigma I_{\text{entering}} = \Sigma I_{\text{leaving}} \Rightarrow I_1 + I_2 = I_3 + I_4 \Rightarrow 4 + 7 = 6 + I_4 \Rightarrow I_4 = 5.0 \text{ A}$$

Fig 1.2a: Kirchhoff's Current Law at Junction Node A (Conservation of Electric Charge)

Step 1 🚀 : State Kirchhoff's Current Law

$$\Sigma I_{\text{in}} = \Sigma I_{\text{out}}$$

Step 2 ⚡ : Substitute Known Values

$$4 + 7 = 6 + I_4 \Rightarrow 11 = 6 + I_4 \Rightarrow \mathbf{I_4 = 5.0, A}$$

I_4 leaves the node with 5.0, A.

🎯 By KCL: Unknown Current $\mathbf{I_4 = 5.0, A}$ (Leaving).

University Question (8 Marks):

A circuit has two meshes:

Mesh 1 has a +30,V battery and resistor $R_1 = 4, \Omega$.

Mesh 2 has a -20,V battery and resistor $R_2 = 6, \Omega$.

The central shared branch has $R_3 = 2, \Omega$ carrying current $(I_1 - I_2)$.

Find loop currents I_1 and I_2 .

SCHEMATIC 2: TWO-MESH KVL CIRCUIT WITH OPPOSING BATTERIES

A

B (Junction)

C

D

E (Return)

F

$$R_1 = 4.0 \, \Omega$$

+

–

+

–

$$+30 \, \text{V}$$

$$R_3 = 2.0 \, \Omega$$

$$I_1 - I_2 \downarrow$$

+

–

$$R_2 = 6.0 \, \Omega$$

+

–

+

-

20 V

(Opposes I_2)

I_1 ↻

Mesh 1 (Clockwise)

I_2 ↻

Mesh 2 (Clockwise)

KVL Mesh 1: $30 - 4 \cdot I_1 - 2(I_1 - I_2) = 0$ | KVL Mesh 2: $-2(I_2 - I_1) - 6 \cdot I_2 - 20 = 0$

Fig 1.2b: Two-Mesh Planar Network with Mesh Current Loops and Central Branch Interaction

Step 1 🚀 : Write KVL for Mesh 1

$$30 - 4I_1 - 2(I_1 - I_2) = 0 \Rightarrow 6I_1 - 2I_2 = 30 \text{— (Eq 1)}$$

Step 2 ⚡ : Write KVL for Mesh 2

$$-2(I_2 - I_1) - 6I_2 - 20 = 0 \Rightarrow -2I_1 + 8I_2 = -20 \text{— (Eq 2)}$$

Step 3 🌀 : Solve Simultaneously

Multiply Eq 2 by 3: $-6I_1 + 24I_2 = -60$.

Add to Eq 1:

$$22I_2 = -30 \Rightarrow I_2 = -1.364, \text{ A}$$

Substitute I_2 into Eq 1:

$$6I_1 - 2(-1.364) = 30 \Rightarrow 6I_1 = 27.272 \Rightarrow I_1 = 4.545, \text{ A}$$



Mesh Currents: $I_1 = 4.55, \text{ A}$ (clockwise), $I_2 = -1.36, \text{ A}$ (counter-clockwise).

University Question (12 Marks - University Exam):

In a bridge network, three mesh currents I_1, I_2, I_3 circulate:

$$\begin{bmatrix} 8 & -2 & -4 \\ -2 & 9 & -3 \\ -4 & -3 & 12 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 24 \\ 0 \\ 0 \end{bmatrix}$$

Find current I_1 and branch current through the shared $4, \Omega$ resistor ($I_1 - I_3$) using Cramer's rule.

A

B

C

D

E

G

+

–

24 V

$R_a = 2.0\ \Omega$

$R_{12} = 2.0\ \Omega$
(Shared M1-M2)

$R_{13} = 4.0\ \Omega$ (Target)

$I_1 - I_3 \downarrow$

$R_b = 4.0\ \Omega$

$R_{23} = 3.0\ \Omega$ (Bridge Arm)

$$R_c = 5.0 \, \Omega$$

$$I_1 \curvearrowright$$

Mesh 1 ($R_{11}=8\Omega$)

$$I_2 \curvearrowright$$

Mesh 2 ($R_{22}=9\Omega$)

$$I_3 \curvearrowright$$

Mesh 3 ($R_{33}=12\Omega$)

MATRIX MAPPING

- $R_{11} = 2+2+4 = 8 \, \Omega$
- $R_{22} = 2+4+3 = 9 \, \Omega$
- $R_{33} = 4+3+5 = 12 \, \Omega$
- $R_{12} = R_{21} = -2 \, \Omega$
- $R_{13} = R_{31} = -4 \, \Omega$
- $R_{23} = R_{32} = -3 \, \Omega$
- $V_s = [24, 0, 0]^T$

Target Current through Shared $4.0 \, \Omega$ Branch: $I_{\{4\Omega\}} = I_1 - I_3 = 4.304 \, \text{A} - 1.826 \, \text{A} = 2.478 \, \text{A}$

Fig 1.2c: 3-Mesh Planar Bridge Circuit Directly Formulating the $[R][I] = [V]$ Impedance Matrix

Step 1 : Evaluate Main Determinant Δ

$$\Delta = 8[(9)(12) - (-3)(-3)] - (-2)[(-2)(12) - (-3)(-4)] + (-4)[(-2)(-3) - (9)(-4)]$$

$$= 8[108 - 9] + 2[-24 - 12] - 4[6 + 36] = 8(99) + 2(-36) - 4(42)$$

$$= 792 - 72 - 168 = \mathbf{552}$$

Step 2 ⚡ : Evaluate Δ_1 for I_1

Replace column 1 with $[24, 0, 0]^T$:

$$\Delta_1 = 24[(9)(12) - (-3)(-3)] = 24(99) = \mathbf{2376}$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{2376}{552} = \mathbf{4.304, \text{ Amperes}}$$

Step 3 🌀 : Evaluate Δ_3 for I_3

Replace column 3 with $[24, 0, 0]^T$:

$$\Delta_3 = 24[(-2)(-3) - (9)(-4)] = 24(6 + 36) = 24(42) = \mathbf{1008}$$

$$I_3 = \frac{\Delta_3}{\Delta} = \frac{1008}{552} = \mathbf{1.826, \text{ Amperes}}$$

Step 4 🧩 : Branch Current

$$I_{4\Omega} = I_1 - I_3 = 4.304 - 1.826 = \mathbf{2.478, \text{ Amperes}}$$

🌀 Mesh Currents: $I_1 = 4.30, \text{ A}, I_3 = 1.83, \text{ A}$ | Branch Current through $4, \Omega$: **2.48, A**.

3. Thevenin's Theorem & Maximum Power (The Ultimate Shortcut)

1. In-Depth Analysis of Thevenin's and Norton's Network Theorems

Network reduction theorems replace complex multi-loop linear networks with simple two-terminal equivalent generators.

A. Thevenin's Theorem (1883):

● **Formal Theorem Statement:** Any linear two-terminal bilateral active network containing independent or dependent sources and linear resistors can be replaced at its external terminals $A - B$ by an equivalent circuit consisting of a single independent ideal voltage source V_{th} in series with an equivalent internal resistance R_{th} .

SCHEMATIC 1: THEVENIN EQUIVALENT GENERATOR DRIVING VARIABLE LOAD R_L

Linear Active Network (Thevenin Equivalent)

+
-
V_{th} (V_{oc})

R_{th}
(Internal Resistance)

$I_L \rightarrow$

Terminal A

Terminal B

R_L (Load)
(Adjustable)

V_L

Loop Current: $I_L = V_{th} / (R_{th} + R_L)$ | Load Power: $P_L = I_L^2 \cdot R_L = (V_{th}^2 \cdot R_L) / (R_{th} + R_L)^2$

Fig 1.3a: Thevenin Equivalent Circuit Connected to Variable Load Resistor R_L

Thevenin Voltage (V_{th}): The open-circuit voltage appearing across terminals $A - B$ when the load resistor R_L is disconnected:

$$V_{th} = V_{oc} = V_A - V_B (\text{with } R_L \rightarrow \infty)$$

Thevenin Resistance (R_{th}): The resistance measured looking back into terminals $A - B$ with all internal independent sources deactivated:

- Independent Voltage Sources $\rightarrow$ **Short-Circuited** ($V = 0$, ∇ , bare conducting wire).
- Independent Current Sources $\rightarrow$ **Open-Circuited** ($I = 0$, Λ , open cut branch).

- **Networks containing Dependent Sources:** You cannot simply combine resistors! Instead, apply an external test source $V_{\text{test}} = 1.0, \text{V}$ across terminals $A - B$, calculate resulting test current I_{test} , and evaluate:

$$R_{\text{th}} = \frac{V_{\text{test}}}{I_{\text{test}}} = \frac{V_{\text{oc}}}{I_{\text{sc}}}$$

B. Norton's Theorem (Dual Generator Representation):

Dual of Thevenin's Theorem: Replaces the active network with an ideal current generator I_N connected in parallel with Norton resistance R_N :

$$I_N = I_{\text{sc}}(\text{short-circuit current flowing between terminals } A \text{ and } B), R_N = R_{\text{th}}$$

$$\text{Source Transformation Duality: } V_{\text{th}} = I_N \cdot R_{\text{th}} \Leftrightarrow I_N = \frac{V_{\text{th}}}{R_{\text{th}}}$$

2. Maximum Power Transfer Theorem (MPTT) Analytical Derivation

University Theorem Statement:

In any linear bilateral DC electrical network delivering energy to a variable load resistor R_L , the active power delivered to the load is maximum when the load resistance equals the internal Thevenin resistance of the driving network looking back from the load terminals:

$$R_L = R_{\text{th}}$$

Step 1 🚀 : Circuit Modeling & Mathematical Expression for Load Power P_L

Connecting an adjustable load resistor R_L across the Thevenin equivalent terminals $A - B$ yields a single-loop series circuit:

By Kirchhoff's Voltage Law (KVL) and Ohm's Law, the loop current is:

$$I_L = \frac{V_{\text{th}}}{R_{\text{th}} + R_L}$$

The active Joule electrical power dissipated in the load resistor is:

$$P_L = I_L^2 \cdot R_L = \left(\frac{V_{th}}{R_{th} + R_L} \right)^2 R_L = \frac{V_{th}^2 \cdot R_L}{(R_{th} + R_L)^2}$$

Step 2 ⚡ : Differential Calculus Optimization (Quotient Rule Derivation)

To find the load resistance R_L that maximizes P_L , we treat P_L as a continuous function of the independent variable R_L while V_{th} and R_{th} remain constant. Applying the Quotient Rule of differential calculus $\frac{d}{dx} \left[\frac{u}{v} \right] = \frac{u'v - uv'}{v^2}$:

$$\begin{aligned} \frac{dP_L}{dR_L} &= V_{th}^2 \cdot \frac{d}{dR_L} \left[\frac{R_L}{(R_{th} + R_L)^2} \right] \\ &= V_{th}^2 \cdot \left[\frac{1 \cdot (R_{th} + R_L)^2 - R_L \cdot 2(R_{th} + R_L)}{(R_{th} + R_L)^4} \right] \\ &= V_{th}^2 \cdot \left[\frac{(R_{th} + R_L) \cdot \{(R_{th} + R_L) - 2R_L\}}{(R_{th} + R_L)^4} \right] \\ &= V_{th}^2 \cdot \frac{R_{th} - R_L}{(R_{th} + R_L)^3} \end{aligned}$$

Step 3 🎯 : First Derivative Test (Condition for Stationary Point)

Setting the first derivative to zero to identify the critical point:

$$\frac{dP_L}{dR_L} = 0 \iff V_{th}^2 \cdot \frac{R_{th} - R_L}{(R_{th} + R_L)^3} = 0$$

Since $V_{th} \neq 0$ and $(R_{th} + R_L)^3 \neq 0$, the numerator must vanish identically:

$$R_{th} - R_L = 0 \Rightarrow \boxed{R_L = R_{th}}$$

Step 4 🔍 : Rigorous Second Derivative Verification ($\frac{d^2 P_L}{dR_L^2} < 0$)

To mathematically guarantee that $R_L = R_{th}$ corresponds to a **global maximum** and not a minimum or inflection point:

$$\begin{aligned}
\frac{d^2 P_L}{dR_L^2} &= V_{th}^2 \cdot \frac{d}{dR_L} \left[\frac{R_{th} - R_L}{(R_{th} + R_L)^3} \right] \\
&= V_{th}^2 \cdot \left[\frac{(-1)(R_{th} + R_L)^3 - (R_{th} - R_L) \cdot 3(R_{th} + R_L)^2}{(R_{th} + R_L)^6} \right] \\
&= V_{th}^2 \cdot \left[\frac{-(R_{th} + R_L) - 3(R_{th} - R_L)}{(R_{th} + R_L)^4} \right] = V_{th}^2 \cdot \frac{2R_L - 4R_{th}}{(R_{th} + R_L)^4}
\end{aligned}$$

Evaluating the second derivative specifically at the critical point $R_L = R_{th}$:

$$\left. \frac{d^2 P_L}{dR_L^2} \right|_{R_L=R_{th}} = V_{th}^2 \cdot \frac{2R_{th} - 4R_{th}}{(2R_{th})^4} = -\frac{2V_{th}^2 R_{th}}{16R_{th}^4} = -\frac{V_{th}^2}{8R_{th}^3} < 0$$

Because $V_{th} > 0$ and $R_{th} > 0$, the second derivative is strictly negative. This **rigorously proves that power P_L reaches a global maximum** when $R_L = R_{th}$!

Step 5 🧩: Maximum Power Delivered to Load

Substituting $R_L = R_{th}$ back into the load power formulation:

$$P_{L,\max} = \left. \frac{V_{th}^2 \cdot R_L}{(R_{th} + R_L)^2} \right|_{R_L=R_{th}} = \frac{V_{th}^2 \cdot R_{th}}{(2R_{th})^2} = \frac{V_{th}^2}{4R_{th}}$$

GRAPH 1: POWER DELIVERED TO LOAD (P_L) VS LOAD RESISTANCE (R_L)

R_L (Ω)

P_L (W)

0

$dP_L / dR_L = 0$ (Peak Maximum)

$R_L = R_{th}$

$P_{L,max}$

$P_{L,max} = V_{th}^2 / (4 \cdot R_{th})$

REGION 1: $R_L < R_{th}$
Current-Limited (High I_L)

REGION 2: $R_L > R_{th}$
Voltage-Limited (Low I_L)

Efficiency at Peak: $\eta = 50\%$ (50% Lost in R_{th})

Fig 1.3b: Maximum Power Transfer Characteristic Curve Demonstrating Global Peak at $R_L = R_{th}$

Step 6 ⚡ : Power Transfer Efficiency Formulation ($\eta = 50$)

At maximum power transfer condition:

Total electrical power generated by the Thevenin voltage source:

$$P_{\text{source}} = I_L \cdot V_{th} = \left(\frac{V_{th}}{2R_{th}}\right) V_{th} = \frac{V_{th}^2}{2R_{th}}$$

Internal Joule heat dissipated inside Thevenin resistance R_{th} :

$$P_{\text{internal}} = I_L^2 \cdot R_{th} = \left(\frac{V_{th}}{2R_{th}}\right)^2 R_{th} = \frac{V_{th}^2}{4R_{th}}$$

Transmission Efficiency (η):

$$\eta = \frac{P_{L,\text{max}}}{P_{\text{source}}} \times 100$$

The remaining 50 of generated power is inevitably burned as Joule waste heat inside the internal resistance R_{th} .

Step 7 🎓 : Crucial University Exam & Viva Insights

| Engineering Field | Telecommunications & Audio Systems | Electrical Power Grid Transmission |

| :--- | :--- | :--- |

| **Primary Objective** | **Maximum Signal Power Transfer** ($R_L = R_{th}$) | **Maximum Transmission Efficiency** ($\eta > 95, R_L \gg R_{th}$) |

| **Physical Rationale** | Signals are milliwatts. Impedance matching eliminates wave reflection and maximizes signal-to-noise ratio. | Grids transmit Gigawatts. Burning 50 power inside generator windings would cause instant thermal meltdown! |

| **AC Conjugate Rule** | For AC circuits ($Z_{th} = R_{th} + jX_{th}$), maximum power occurs when load is the complex conjugate: $Z_L = Z_{th}^* = R_{th} - jX_{th}$. | Generators operate with internal impedance $Z_g \approx 0$ relative to network load. |

Problem:

A circuit has a 20, V DC source connected to a voltage divider consisting of $R_1 = 6, \Omega$ and $R_2 = 4, \Omega$. A load $R_L = 5, \Omega$ is connected across R_2 . Find V_{th} , R_{th} , and load current I_L .

GIVEN SCHEMATIC: 20V SOURCE WITH VOLTAGE DIVIDER (6Ω, 4Ω) & LOAD R_L = 5Ω

+

−

20 V

$$R_1 = 6.0 \, \Omega$$

$$R_2 = 4.0 \, \Omega$$

Terminal A

Terminal B

$$R_L = 5.0 \, \Omega$$

Given: 20V supply, $R_1 = 6 \, \Omega$, $R_2 = 4 \, \Omega$, $R_L = 5 \, \Omega \Rightarrow$ Find Thevenin (V_{th} , R_{th}) and Load Current I_L

Fig 1.3c: Given Voltage Divider Circuit with Connected Load Resistor R_L

Step 1 🚀 : Find V_{th} (Open-circuit voltage across R_2)

$$V_{th} = 20 \times \frac{4}{6+4} = 20 \times 0.4 = \mathbf{8.0, V}$$

Step 2 ⚡ : Find R_{th} (Deactivate 20V battery with short circuit)

$$R_{th} = R_1 \parallel R_2 = \frac{6 \times 4}{6 + 4} = \frac{24}{10} = \mathbf{2.4, \Omega}$$

Step 3 🎯: Load Current I_L

$$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{8.0}{2.4 + 5.0} = \frac{8.0}{7.4} = \mathbf{1.08, A}$$

SOLVED SCHEMATIC: THEVENIN EQUIVALENT CIRCUIT DRIVING 5Ω LOAD

+

–

V_{th}=8.0V

$$R_{th} = 2.4 \, \Omega$$

$$I_L = 1.08 \, A$$

A

B

$$R_L = 5.0 \, \Omega$$

Solved: $V_{th} = 8.0 \, V$ | $R_{th} = 2.4 \, \Omega$ | $I_L = 8.0 / (2.4 + 5.0) = 1.08 \, \text{Amperes}$

Fig 1.3d: Solved Single-Loop Thevenin Circuit Powering R_L

🌀 $V_{th} = 8.0, V$ | $R_{th} = 2.4, \Omega$ | Load Current $I_L = 1.08, A$.

University Question (10 Marks):

A bridge circuit has a 60,V battery connected to two parallel legs:

Branch 1: 10,\Omega in series with 20,\Omega. Node A is between them.

Branch 2: 30,\Omega in series with 15,\Omega. Node B is between them.

Find V_{th} , R_{th} , and maximum power delivered to a variable load R_L .

GIVEN SCHEMATIC: 60V BRIDGE CIRCUIT (BRANCH 1: 10Ω-20Ω | BRANCH 2: 30Ω-15Ω)

+

-

60 V

$R_1 = 10\ \Omega$

A (40V)

$R_2 = 20\ \Omega$

$$R_3 = 30 \, \Omega$$

$$B \, (20V)$$

$$R_4 = 15 \, \Omega$$

$$R_L \, (\text{Load})$$

Given: $V = 60V$, Branch 1 ($10\Omega, 20\Omega$), Branch 2 ($30\Omega, 15\Omega$) $\Rightarrow$ Find V_{th} , R_{th} , and P_{max}

Fig 1.3e: Given Bridge Circuit Network with Load R_L Across Terminals A-B

Step 1 🚀 : Find Thevenin Voltage $V_{th} = V_A - V_B$

$$V_A = 60 \times \frac{20}{10 + 20} = 40.0, V, V_B = 60 \times \frac{15}{30 + 15} = 20.0, V$$

$$V_{th} = 40.0 - 20.0 = \mathbf{20.0, V}$$

Step 2 ⚡ : Find Thevenin Resistance R_{th}

$$R_{th} = (10 \parallel 20) + (30 \parallel 15) = \frac{200}{30} + \frac{450}{45} = 6.67 + 10.0 = \mathbf{16.67, \Omega}$$

Step 3 🎯 : Maximum Power Transfer Condition & Value

By MPTT, set $R_L = R_{th} = 16.67, \Omega$.

$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{(20)^2}{4(16.67)} = \frac{400}{66.67} = \mathbf{6.0, Watts}$$

SOLVED SCHEMATIC: THEVENIN EQUIVALENT ($V_{th} = 20V$, $R_{th} = 16.67\Omega$) DELIVERING P_{MAX}

+

-

$V_{th}=20V$

$R_{th} = 16.67 \Omega$

A

B

$$R_L = R_{th} = 16.67 \, \Omega$$

$$\text{Solved: } V_{th} = 20.0 \, \text{V} \mid R_{th} = 16.67 \, \Omega \mid P_{max} = (20)^2 / (4 \times 16.67) = 6.00 \, \text{Watts}$$

Fig 1.3f: Solved Thevenin Equivalent Circuit Powering Optimal Load (MPTT)

 $V_{th} = 20.0, \text{V} \mid R_{th} = 16.67, \Omega \mid \text{Optimal } R_L = 16.67, \Omega \mid P_{max} = 6.0, \text{W}.$

University Question (12 Marks - Exam Challenge):

Find the Thevenin equivalent looking into terminals A-B for a circuit containing an independent 10V source, a $4 \, \Omega$ resistor carrying control current i_x , a dependent voltage source $2 i_x$, and a $6 \, \Omega$ output resistor.

GIVEN SCHEMATIC: 10V INDEPENDENT SOURCE, $4 \, \Omega$ (WITH CURRENT i_x) & DEPENDENT SOURCE $2 i_x$

–

10 V

$i_x \rightarrow$

$4.0 \, \Omega$

+

–

$2 \cdot i_x$

$6.0 \, \Omega$

Terminal A

Terminal B

Given: Independent 10V source, 4Ω with control current i_x , dependent $2 \cdot i_x$ source, 6Ω load branch

Fig 1.3g: Dependent Source Circuit Schematic with Control Current i_x

Step 1 🚀 : Find Open-Circuit Voltage $V_{oc} = V_{th}$

With terminals A-B open:

Loop current: $i_x = \frac{10-2i_x}{4+6} \Rightarrow 10i_x = 10 - 2i_x \Rightarrow 12i_x = 10 \Rightarrow i_x = \frac{5}{6}, \text{A}.$

Output voltage across $6, \Omega$:

$$V_{th} = i_x \times 6 = \frac{5}{6} \times 6 = \mathbf{5.0, \text{Volts}}$$

Step 2 ⚡ : Find R_{th} using Test Source Method ($V_{test} = 1, \text{V}$)

Deactivate independent source ($10, \text{V} \rightarrow 0, \text{V}$ short circuit). Keep dependent source $2i_x$ active!

Apply test source $V_{test} = 1.0, \text{V}$ at terminals A-B:

From node equation: i_x flows in reverse: $i_x = -\frac{V_{test}+2i_x}{4} \Rightarrow 4i_x = -1 - 2i_x \Rightarrow 6i_x = -1 \Rightarrow i_x = -1/6, \text{A}.$

Current from test generator:

$$I_{test} = \frac{V_{test}}{6} - i_x = \frac{1}{6} - (-\frac{1}{6}) = \frac{2}{6} = \frac{1}{3}, \text{A}$$

Looking-back resistance:

$$R_{th} = \frac{V_{test}}{I_{test}} = \frac{1.0}{1/3} = \mathbf{3.0, \Omega}$$

SOLVED SCHEMATIC: DEPENDENT SOURCE THEVENIN EQUIVALENT ($V_{th} = 5.0\text{V}$, $R_{th} = 3.0\Omega$)

+

-

$$V_{th}=5.0V$$

$$R_{th} = 3.0 \, \Omega$$

Terminal A

Terminal B

Test Source Result

$$V_{test} = 1.0 \, V$$

$$I_{test} = 1/3 \, A \Rightarrow R_{th} = 3\Omega$$

Solved: Open-circuit $V_{th} = 5.0 \, V$ | Test generator method: $R_{th} = V_{test} / I_{test} = 3.0 \, \Omega$

Fig 1.3h: Solved Thevenin Equivalent with Dependent Source Parameters

 Dependent Source Thevenin: $V_{th} = 5.0, V$ | $R_{th} = 3.0, \Omega$.

4. AC Waveforms: RMS, Average & Phase (Why Household Sockets Say 220V)

1. Alternating Current (AC) Sinusoidal Waveform Theory

An alternating quantity is one whose magnitude and direction reverse periodically with time. The mathematical expression of a sinusoidal voltage wave is:

$$v(t) = V_m \sin(\omega t \pm \phi)$$

V_m (**Amplitude / Crest Value**): Peak positive excursion from zero.

$\omega = 2\pi f = \frac{2\pi}{T}$: Angular frequency in radians per second (rad/s).

$f = 1/T$: Cyclic frequency in Hertz (Hz or cycles per second).

ϕ : Initial phase angle relative to time origin $t = 0$.

2. Analytical Derivation of RMS and Average Values

A. Root Mean Square (RMS) Value (Effective Value):

● **Physical Definition:** The RMS value of an alternating current is defined as that value of steady direct current (DC) which, when flowing through a given resistor for a given time, produces the **exact same quantity of Joule heat** as the AC does in the same resistor over one full period T .

$$H_{dc} = I_{rms}^2 RT \text{ and } H_{ac} = \int_0^T [i(t)]^2 R, dt$$

Equating:

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^T [i(t)]^2, dt}$$

Mathematical Derivation for Pure Sinusoid $i(t) = I_m \sin \theta$:

$$I_{rms}^2 = \frac{1}{2\pi} \int_0^{2\pi} I_m^2 \sin^2 \theta, d\theta = \frac{I_m^2}{2\pi} \int_0^{2\pi} \frac{1 - \cos 2\theta}{2} d\theta = \frac{I_m^2}{4\pi} [\theta]_0^{2\pi} = \frac{I_m^2}{4\pi} (2\pi) = \frac{I_m^2}{2}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}} \approx 0.7071 \cdot I_m, V_{rms} = \frac{V_m}{\sqrt{2}} \approx 0.7071 \cdot V_m$$

B. Average Value (I_{avg}):

For a symmetrical wave, average over a full cycle is zero. Hence, the average value is defined over **one positive half-cycle**:

$$I_{avg} = \frac{1}{\pi} \int_0^{\pi} I_m \sin \theta, d\theta = \frac{I_m}{\pi} [-\cos \theta]_0^{\pi} = \frac{I_m}{\pi} [1 - (-1)] = \frac{2I_m}{\pi} \approx 0.6366 \cdot I_m$$

3. Waveform Shape Characteristics

Form Factor (K_f): Ratio of RMS value to Average value:

$$K_f = \frac{V_{rms}}{V_{avg}} = \frac{V_m / \sqrt{2}}{2I_m / \pi} = \frac{\pi}{2\sqrt{2}} \approx 1.11$$

Peak / Crest Factor (K_p): Ratio of Peak value to RMS value:

$$K_p = \frac{V_m}{V_{rms}} = \frac{V_m}{V_m / \sqrt{2}} = \sqrt{2} \approx 1.414$$

Problem:

A voltage wave is represented by $v(t) = 141.4 \sin(314t)$, V. Find: (1) Peak voltage V_m , (2) RMS voltage V_{rms} , (3) Frequency f , (4) Time period T .

Step 1 🚀 : Compare with Standard Equation $v(t) = V_m \sin(\omega t)$

$$V_m = 141.4, \text{ Volts}$$

Step 2 ⚡ : Calculate RMS Voltage

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{141.4}{1.414} = 100.0, \text{ Volts}$$

Step 3 🎯 : Frequency & Time Period

$$\omega = 2\pi f = 314 \Rightarrow f = \frac{314}{2\pi} = \frac{314}{6.283} = \mathbf{50.0, Hz}$$

$$T = \frac{1}{f} = \frac{1}{50} = \mathbf{0.020, s = 20, ms}$$

🎯 $V_m = 141.4, V \mid V_{rms} = 100.0, V \mid$ Frequency $f = 50.0, Hz \mid$ Period $T = 20.0, ms$.

University Question (6 Marks):

Calculate the average value, form factor, and crest factor of an AC sinusoidal current of peak amplitude $I_m = 20, A$.

Step 1 🚀 : Average Value (Over Half Cycle)

$$I_{avg} = (2 I_m / \pi) = (2 \times 20 / \pi) = (40 / 3.1416) = 12.73, A$$

Step 2 ⚡ : RMS Value

$$I_{rms} = (I_m / \sqrt{2}) = (20 / 1.4142) = 14.14, A$$

Step 3 🎯 : Form Factor (Kf) & Peak / Crest Factor (Kp)

$$K_f = (I_{rms} / I_{avg}) = (14.14 / 12.73) = 1.11$$

$$K_p = (I_m / I_{rms}) = (20 / 14.14) = 1.414 = \sqrt{2}$$

🎯 $I_{avg} = 12.73, A \mid I_{rms} = 14.14, A \mid$ Form Factor $K_f = 1.11 \mid$ Crest Factor $K_p = 1.414$.

University Question (10 Marks):

Derive from first principles the RMS and Average values of a half-wave rectified sinusoidal voltage defined by:

$$v(\theta) = \begin{cases} V_m \sin \theta, & 0 \leq \theta \leq \pi \\ 0, & \pi \leq \theta \leq 2\pi \end{cases}$$

Hence find its Form Factor.

Step 1 🚀 : Average Value V_{avg}

$$V_{avg} = \frac{1}{2\pi} \int_0^\pi V_m \sin \theta d\theta = \frac{V_m}{2\pi} [-\cos \theta]_0^\pi = \frac{V_m}{2\pi} (1 - (-1)) = \frac{V_m}{\pi} \approx \mathbf{0.318 V_m}$$

Step 2 ⚡ : RMS Value V_{rms}

$$V_{rms}^2 = \frac{1}{2\pi} \int_0^\pi V_m^2 \sin^2 \theta d\theta = \frac{V_m^2}{2\pi} \int_0^\pi \frac{1 - \cos 2\theta}{2} d\theta = \frac{V_m^2}{4\pi} [\theta]_0^\pi = \frac{V_m^2}{4}$$

$$V_{rms} = \sqrt{\frac{V_m^2}{4}} = \frac{V_m}{2} = 0.50 V_m$$

Step 3 🎯 : Form Factor (K_f)

$$K_f = \frac{V_{rms}}{V_{avg}} = \frac{V_m/2}{V_m/\pi} = \frac{\pi}{2} \approx 1.57$$

🎯 Half-Wave: $V_{avg} = \frac{V_m}{\pi}$ | $V_{rms} = \frac{V_m}{2}$ | Form Factor $K_f = 1.57$ (High distortion compared to pure sine 1.11).

5. Series R-L-C Circuit & Electrical Resonance (The Radio Tuning Magic)

1. Complex Impedance & Series R-L-C Network Analysis

In the frequency domain (phasor representation), circuit elements are characterized by complex impedances:

Resistor (R): $Z_R = R \angle 0^\circ = R + j0$. Voltage and current are strictly in phase ($\phi = 0^\circ$).

Inductor (L): $Z_L = j\omega L = \omega L \angle +90^\circ$. Inductive reactance $X_L = 2\pi fL$, $[\Omega]$. Voltage leads current by $+90^\circ$.

Capacitor (C): $Z_C = \frac{1}{j\omega C} = -j \frac{1}{\omega C} = \frac{1}{\omega C} \angle -90^\circ$. Capacitive reactance $X_C = \frac{1}{2\pi fC}$, $[\Omega]$. Voltage lags current by -90° .

Total Series Impedance:

$$\mathbf{Z} = \mathbf{R} + \mathbf{j}(\mathbf{X}_L - \mathbf{X}_C) = |\mathbf{Z}| \angle \phi$$

$$|\mathbf{Z}| = \sqrt{R^2 + (X_L - X_C)^2}, \phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

2. Theoretical Analysis of Series Electrical Resonance

● **Resonance Definition:** Series resonance occurs in an R-L-C circuit at the particular frequency where the net reactive component of the total impedance vanishes identically:

$$\text{Im}(\mathbf{Z}) = \mathbf{X}_L - \mathbf{X}_C = \mathbf{0} \Leftrightarrow \mathbf{X}_L = \mathbf{X}_C$$

Derivation of Resonant Frequency (f_0):

$$2\pi f_0 L = \frac{1}{2\pi f_0 C} \Rightarrow f_0^2 = \frac{1}{4\pi^2 LC} \Rightarrow f_0 = \frac{1}{2\pi \sqrt{LC}}, \omega_0 = \frac{1}{\sqrt{LC}}$$

Phenomenological Characteristics at Resonance:

Minimum Impedance: $Z_{\min} = R$ (purely resistive, real).

Maximum Current (Acceptor Circuit): $I_0 = \frac{V}{Z_{\min}} = \frac{V}{R}$.

Unity Power Factor: $\cos \phi = \frac{R}{|Z|} = \frac{R}{R} = 1.0$.

Voltage Magnification: Although net reactive voltage is zero ($V_L + V_C = 0$), the individual voltages across inductor and capacitor can reach destructive levels:

$$V_L = I_0 X_L = \left(\frac{V}{R}\right)(\omega_0 L) = Q \cdot V, V_C = I_0 X_C = Q \cdot V$$

3. Quality Factor (Q), Bandwidth & Selectivity

The **Quality Factor (Q)** measures the sharp selectivity of the circuit:

$$Q = \frac{\text{Reactive Power}}{\text{Active Power}} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Half-Power Frequencies (f_1, f_2): Frequencies where current drops to $I_0 / \sqrt{2}$ (70.7) and power drops by 50.

Bandwidth (BW):

$$BW = \Delta f = f_2 - f_1 = \frac{f_0}{Q} = \frac{R}{2\pi L}$$

Problem:

A coil has resistance $R = 30, \Omega$ and inductive reactance $X_L = 40, \Omega$ across a 200, V, 50, Hz supply. Find total impedance Z , current I , and power factor $\cos \phi$.

Step 1 🚀 : Impedance Magnitude

$$|Z| = \sqrt{R^2 + X_L^2} = \sqrt{30^2 + 40^2} = \sqrt{900 + 1600} = \sqrt{2500} = 50.0, \Omega$$

Step 2 ⚡ : Current Magnitude

$$I = \frac{V}{|Z|} = \frac{200, V}{50, \Omega} = 4.0, \text{Amperes}$$

Step 3 🎯 : Power Factor

$$\cos \phi = \frac{R}{|Z|} = \frac{30}{50} = 0.60, \text{lagging}$$

🎯 Impedance $Z = 50.0, \Omega$ | Current $I = 4.0, A$ | Power Factor = 0.60 (lagging).

University Question (8 Marks):

A coil of $R = 10, \Omega$, $L = 0.15, H$, and $C = 100, \mu F$ is connected across 230, V, 50, Hz.

Find: (1) Total impedance at 50, Hz, (2) Resonant frequency f_0 , (3) Peak current at resonance I_0 .

Step 1 🚀 : Reactances at 50, Hz

$$X_L = 2\pi(50)(0.15) = 47.12, \Omega, \quad X_C = \frac{1}{2\pi(50)(100 \times 10^{-6})} = 31.83, \Omega$$

$$|Z| = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{10^2 + (47.12 - 31.83)^2} = \sqrt{100 + 233.78} = 18.27, \Omega$$

Step 2 ⚡ : Resonant Frequency f_0

$$f_0 = \frac{1}{2\pi LC} = \frac{1}{2\pi \cdot 0.15 \times 100 \times 10^{-6}} = \frac{1}{2\pi(0.00387)} = 41.09, \text{Hz}$$

Step 3 🎯 : Current at Resonance

$$\text{At resonance: } X_L = X_C \Rightarrow Z_{min} = R = 10, \text{Omega}$$

$$I_0 = \frac{V}{R} = \frac{230}{10} = 23.0, \text{Amperes}$$

🎯 At 50Hz: $Z = 18.27, \Omega$ | Resonant Frequency $f_0 = 41.09, \text{Hz}$ | Resonant Current $I_0 = 23.0, \text{A}$.

University Question (10 Marks - University Exam):

A series resonant circuit with $R = 5, \Omega$, $L = 0.2, \text{H}$, and $C = 5, \mu\text{F}$ is connected to a $50, \text{V}$ variable frequency supply.

Determine: (1) Resonant frequency f_0 , (2) Quality factor Q , (3) Bandwidth BW , (4) Voltage across the capacitor V_C at resonance.

Step 1 🎯 : Resonant Frequency

$$f_0 = \frac{1}{2\pi\sqrt{(0.2 \times 5 \times 10^{-6})}} = \frac{1}{2\pi\sqrt{(10^{-6})}} = \frac{1000}{2\pi} = 159.15, \text{Hz}$$

$$\omega_0 = 2\pi f_0 = 1000, \text{ad/s}$$

Step 2 ⚡ : Quality Factor (Q)

$$Q = (\omega_0 L / R) = ((1000)(0.2) / 5) = \frac{200}{5} = 40.0$$

Step 3 🎯 : Bandwidth (BW)

$$BW = (f_0 / Q) = (159.15 / 40) = 3.98, \text{Hz}$$

Step 4 🎯 : Voltage Magnification Across Capacitor at Resonance

At resonance, circuit current is $I_0 = (V / R) = \frac{50}{5} = 10.0, \text{A}$.

Capacitive reactance at resonance:

$$X_C = (1 / \omega_0 C) = (1 / (1000)(5 \times 10^{-6})) = 200, \Omega$$

$$V_C = I_0 \times X_C = 10.0 \times 200 = 2000.0, \text{Volts} = Q \times V_{\text{supply}} = 40 \times 50 = 2000, \text{V}$$

(The voltage across the capacitor is 40 times higher than the input voltage!)

🎯 $f_0 = 159.15, \text{Hz}$ | Quality Factor $Q = 40.0$ | Bandwidth $BW = 3.98, \text{Hz}$ | Resonant Capacitor Voltage $V_C = 2000, \text{V}$ (40x Magnification).

University Solved Examination Questions (Unit 1)

MMDU Official University Solved Examination Questions (Unit 1)

Course: Basic Electrical & Electronics Engineering ('BELE-001')

Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (6 Marks In-Depth), D (6 Marks Design/Long)

EASY

Question 1 (1 Mark): Physical Meaning of Power Factor

Question:

Explain why low power factor is detrimental to industrial electrical transmission networks.



Question Schematic / State Specification:

...

Question Power Triangle: Apparent Power S (kVA) = Active Power P (kW) + j Reactive Power Q (kVAR).

...

Step-by-Step Solution:



Solution Analytical Derivation & Schematic:

...

Solution Current Vector Magnification:

$I = P / (V \cos\phi)$. As $\cos\phi \downarrow \Rightarrow I \uparrow \Rightarrow I^2 R$ Losses $\uparrow$.

...

For a fixed real power P , drawing at low power factor forces current I to increase inversely. This magnifies $I^2 R$ transmission line heating losses, induces excessive line voltage drops, and demands larger conductor cross-sections and transformer ratings.



Verified Result: Question 1 step-by-step resolution per MMDU university marking scheme.

EASY

Question 2 (1 Mark): Resonance Condition in Series RLC Circuit

Question:

State why the net impedance of a series RLC circuit is purely resistive at resonant frequency.

Question Schematic / State Specification:

...

Question Reactance Cancellation: Inductive Reactance $+jX_L$ balances Capacitive Reactance $-jX_C$.

...

Step-by-Step Solution:

Solution Analytical Derivation & Schematic:

...

Solution Phasor Cancellation Model:

$Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$. At $\omega_0 = 1/\sqrt{LC} \Rightarrow X_L = X_C \Rightarrow Z = R + j0 = R$.

...

At resonance, the counter-phase voltages across the inductor and capacitor cancel each other exactly ($V_L = -V_C$). The network behaves as a pure resistance with minimum impedance and unity power factor.



Verified Result: Question 2 step-by-step resolution per MMDU university marking scheme.

EASY

Question 3 (1 Mark): Maximum Power Transfer Efficiency Paradox

Question:

Why is the Maximum Power Transfer Theorem not applied to commercial electrical power utility grids?

Question Schematic / State Specification:

...

Question Efficiency vs Transfer: Utility Goal = 95–98% Efficiency | MPT Goal = Maximum Watts to Load (50% Efficiency).

...

Step-by-Step Solution:

Solution Analytical Derivation & Schematic:

...

Solution Energy Loss Formulation:

Efficiency $\eta = P_L / (P_L + P_{int}) = R_L / (R_L + R_{th}) = 50\%$ when $R_L = R_{th}$.

...

Operating a utility grid at maximum power transfer wastes 50% of generated power as internal heat inside generators and transmission lines. Power systems operate at maximum efficiency ($R_L \gg R_{internal}$) rather than maximum power transfer.



Verified Result: Question 3 step-by-step resolution per MMDU university marking scheme.

MEDIUM

Question 13 (2 Marks): Thevenin Equivalent Circuit Derivation

Question:

Find the Thevenin equivalent circuit across terminals A-B for a circuit with a 24V DC source in series with 6Ω , connected across a parallel 12Ω resistor.

Question Schematic / State Specification:

...

Question Voltage Divider Topology: 24V source $\rightarrow$ 6Ω resistor $\rightarrow$ Node A [12Ω connected to Node B/ground].

...

Step-by-Step Solution:

Solution Analytical Derivation & Schematic:

...

Solution Thevenin Derivation Equations:

$$V_{th} = V_{oc} = 24 \times [12 / (6 + 12)] = 24 \times \frac{12}{18} = 16 \text{ V}$$

$$R_{th} = 6 \parallel 12 = \frac{6 \times 12}{6 + 12} = 72 / 18 = 4 \Omega$$

...

The entire circuit across terminals A-B simplifies to a Thevenin voltage source $V_{th} = 16 \text{ V}$ in series with a resistance $R_{th} = 4 \Omega$.



Verified Result: Question 13 step-by-step resolution per MMDU university marking scheme.

HARD**Question 17 (4 Marks): Complete AC Series RLC Circuit Analysis****Question:**

A coil of resistance $R = 10\Omega$ and inductance $L = 0.1\text{H}$ is connected in series with capacitor $C = 50\mu\text{F}$ across a 230V, 50Hz AC supply. Calculate: (1) Inductive & capacitive reactance, (2) Total impedance, (3) Current and power factor, (4) Active and reactive power.

 Question Schematic / State Specification:

...

Question Series RLC Schematic: 230V, 50Hz $\rightarrow$ R (10Ω) $\rightarrow$ L (0.1H) $\rightarrow$ C ($50\mu\text{F}$).

...

Step-by-Step Solution: **Solution Analytical Derivation & Schematic:****Step-by-Step Impedance & Power Calculations:**

- 1. Reactances:**

$$X_L = 2\pi fL = 2 \times 3.1416 \times 50 \times 0.1 = \mathbf{31.42, \text{Omega}}$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2 \times 3.1416 \times 50 \times (50 \times 10^{-6})} = \mathbf{63.66, \text{Omega}}$$

$$\text{Net Reactance: } X = X_L - X_C = 31.42 - 63.66 = \mathbf{-32.24, \text{Omega} \quad (\text{Capacitive!})}$$

- 2. Circuit Impedance:**

$$|Z| = \sqrt{R^2 + X^2} = \sqrt{10^2 + (-32.24)^2} = \sqrt{100 + 1039.4} = 33.76, \text{ Omega}$$

- **3. Circuit Current:**

$$I = \frac{V}{|Z|} = \frac{230}{33.76} = 6.813, \text{ A}$$

- **4. Power Factor & Power Calculations:**

$$\text{Power Factor: } \cos \phi = \frac{R}{|Z|} = \frac{10}{33.76} = 0.2962 \quad (\text{Leading})$$

$$\text{Active Power: } P = VI \cos \phi = 230 \times 6.813 \times 0.2962 = 464.15, \text{ W}$$

$$\text{Reactive Power: } Q = VI \sin \phi = 230 \times 6.813 \times \sqrt{1 - 0.2962^2} = 1494.5, \text{ VAR (Leading)}$$

Because capacitive reactance dominates ('XC > XL'), the circuit draws a leading current of 6.81 A at a power factor of 0.296 leading.

 **Verified Result:** Question 17 step-by-step resolution per MMDU university marking scheme.

HARD

Question 21 (6 Marks): Superposition and Thevenin Verification on Multi-Source Network

Question:

For a network with two DC voltage sources $V_1 = 30\text{V}$ (with internal resistance 2Ω) and $V_2 = 20\text{V}$ (with internal resistance 4Ω) connected in parallel across a common load resistor $R_L = 8\Omega$. Find current through R_L using: (1) Superposition Theorem, (2) Thevenin's Theorem. Verify both methods give identical results.



Question Schematic / State Specification:

...

Question Parallel Source Bridge Topology: Branch 1 (30V, 2Ω) $\parallel$ Branch 2 (20V, 4Ω) $\rightarrow$ Load R_L (8Ω).

...

Step-by-Step Solution:



Solution Analytical Derivation & Schematic:

...

Solution Step-by-Step Dual Theorem Verification:

`Method 1: Superposition Theorem`

`Case A:  $V_1$  (30V) active,  $V_2$  shorted:`

$$R_{eq,1} = 2 + (4 \parallel 8) = 2 + \frac{32}{12} = 2 + 2.667 = 4.667 \Omega \Rightarrow I_1 = 30 / 4.667 = 6.429 \text{ A}$$

$$\text{Current through } R_L: IL_1 = 6.429 \times [4 / (4 + 8)] = 6.429 \times \frac{4}{12} = 2.143 \text{ A (↓)}`$$

`Case B:  $V_2$  (20V) active,  $V_1$  shorted:`

$$R_{eq,2} = 4 + (2 \parallel 8) = 4 + \frac{16}{10} = 5.60 \Omega \Rightarrow I_2 = 20 / 5.60 = 3.571 \text{ A}$$

$$\text{Current through } R_L: IL_2 = 3.571 \times [2 / (2 + 8)] = 3.571 \times \frac{2}{10} = 0.714 \text{ A (↓)}`$$

$$\text{Total Load Current: } IL = IL_1 + IL_2 = 2.143 + 0.714 = 2.857 \text{ A}$$

`Method 2: Thevenin's Theorem`

$$\text{Open circuit terminals of } R_L: V_{th} = V_2 + (V_1 - V_2) \times [4 / (2 + 4)] = 20 + 10 \times \frac{4}{6} = 26.67 \text{ V}$$

$$R_{th} = 2 \parallel 4 = \frac{2 \times 4}{2 + 4} = 8 / 6 = 1.333 \Omega$$

`Load Current:  $I_L = V_{th} / (R_{th} + R_L) = 26.67 / (1.333 + 8) = 26.67 / 9.333 = 2.857 \text{ A}$ `
```

Both Superposition and Thevenin theorems converge to an identical load current of **2.857 A**.

 **Verified Result:** Question 21 step-by-step resolution per MMDU university marking scheme.

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# Transformers & Electrical Machines (The Magic Gearbox)

**Exam Weightage:** 24 - 28 Marks • **Estimated Study Time:** 35 mins

*Transformers explained as magnetic gearboxes! Step-up vs Step-down, EMF equation derivation, Core vs Copper losses, OC/SC tests explained simply, DC Motor vs Generator working, and Fleming's Left & Right Hand Rules.*

## 1. Single-Phase Transformer Working & EMF Equation

### 1. Electromagnetic Principles of the Single-Phase Transformer

A **transformer** is a static electromagnetic device that transfers electrical energy between two or more circuits via mutual magnetic induction at constant frequency.

#### Working Principle & Faraday's Law:

When sinusoidal voltage  $v_1(t) = V_{m1} \sin(\omega t)$  is connected across primary winding of  $N_1$  turns, an alternating magnetizing current flows, producing sinusoidal core flux:

$$\phi(t) = \phi_m \sin(\omega t) [\text{Weber, Wb}]$$

This common mutual flux links both primary and secondary windings through the high-permeability laminated ferromagnetic core.

By Faraday's Law of Induction, counter-EMFs are induced in both windings:

$$e_1 = -N_1 \frac{d\phi}{dt}, e_2 = -N_2 \frac{d\phi}{dt}$$

### 2. Rigorous Derivation of the Transformer EMF Equation

Substitute sinusoidal flux  $\phi(t) = \phi_m \sin(\omega t)$ :

$$e_1 = -N_1 \frac{d}{dt} [\phi_m \sin(\omega t)] = -N_1 \omega \phi_m \cos(\omega t)$$

Using the trigonometric identity  $-\cos \theta = \sin(\theta - 90^\circ)$ :

$$e_1 = N_1 (2\pi f) \phi_m \sin(\omega t - 90^\circ)$$

The maximum (peak) induced EMF is:

$$E_{m1} = 2\pi f N_1 \phi_m$$

The Root Mean Square (RMS) value is:

$$E_1 = \frac{E_{m1}}{2} = \frac{2\pi}{2} f N_1 \phi_m = 2\pi f N_1 \phi_m$$

Since  $2\pi \approx 1.4142 \times 3.1416 \approx 4.44288$ :

$$\mathbf{E_1 = 4.44 \cdot f \cdot N_1 \cdot \phi_m}$$

Similarly, for the secondary winding:

$$\mathbf{E_2 = 4.44 \cdot f \cdot N_2 \cdot \phi_m}$$

#### EMF per Turn (Core Invariant):

Dividing by the turn count yields the fundamental invariant of transformer design:

$$\frac{\mathbf{E_1}}{\mathbf{N_1}} = \frac{\mathbf{E_2}}{\mathbf{N_2}} = \mathbf{4.44 \cdot f \cdot \phi_m = 4.44 \cdot f \cdot B_m \cdot A_i}$$

where  $B_m$  is maximum core flux density (Tesla) and  $A_i$  is net iron cross-sectional area (m<sup>2</sup>).

---

### 3. Transformation Ratio ( $K$ ) & Impedance Reflection

$$K = \frac{V_2}{V_1} \approx \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$$

$K > 1 \Rightarrow$  **Step-Up Transformer** ( $V_2 > V_1, I_2 < I_1$ ).

$K < 1 \Rightarrow$  **Step-Down Transformer** ( $V_2 < V_1, I_2 > I_1$ ).

#### Resistance and Reactance Referral Rules:

Transferring secondary impedance to primary side:

$$R'_2 = \frac{R_2}{K^2}, x'_2 = \frac{x_2}{K^2}, Z_{eq1} = R_1 + R'_2 + j(x_1 + x'_2)$$

Transferring primary impedance to secondary side:

$$R''_1 = K^2 R_1, x''_1 = K^2 x_1, Z_{eq2} = R_2 + R''_1 + j(x_2 + x''_1)$$

---

#### Problem:

A single-phase transformer has 500 primary turns and 50 secondary turns. If the primary is connected to a 230, V AC source, calculate the secondary voltage  $V_2$  and transformation ratio  $K$ .

GIVEN SCHEMATIC: SINGLE-PHASE STEP-DOWN TRANSFORMER ( $N_1 = 500$ ,  $N_2 = 50$ ,  $V_1 = 230V$ )

$$V_1 = 230 \text{ V (AC)}$$

$$N_1 = 500T$$

CRGO Core

$$N_2 = 50T$$

A

B

$$V_2 = ? \text{ Volts}$$

Given:  $V_1 = 230V$ ,  $N_1 = 500$  turns,  $N_2 = 50$  turns  $\Rightarrow$  Find Transformation Ratio  $K$  and Secondary Voltage  $V_2$

Fig 2.1a: Schematic Representation of Single-Phase Step-Down Core Transformer

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**Step 1 🚀 : Transformation Ratio  $K$**

$$K = \frac{N_2}{N_1} = \frac{50}{500} = \mathbf{0.10} \text{ (Step-down transformer)}$$

## Step 2 ⚡ : Secondary Voltage

$$V_2 = K \cdot V_1 = 0.10 \times 230, \text{ V} = \mathbf{23.0, \text{ Volts}}$$

SOLVED SCHEMATIC: VOLTAGE TRANSFORMATION (K = 0.10,  $V_2 = 23.0\text{V}$  STEPPED-DOWN)

$$V_1 = 230 \text{ V}$$

$$500\text{T}$$

$$\Phi_{\text{core}}$$

$$50\text{T}$$

$$V_2 = 23.0 \text{ Volts}$$

Step-Down ( $K = 0.1$ )

$$\text{Solved: } K = N_2 / N_1 = 50/500 = 0.10 \mid \text{Secondary Voltage: } V_2 = K \cdot V_1 = 0.10 \times 230\text{V} = 23.0 \text{ Volts}$$

Fig 2.1b: Solved Single-Phase Transformer Core Model & Voltage Transformation

🌀 Transformation Ratio  $K = 0.10$  | Secondary Voltage  $V_2 = 23.0, \text{ V}$ .

### University Question (8 Marks):

A 2200 / 220,V, 50,Hz single-phase step-down transformer has 60 turns on the secondary winding. Calculate: (1) Primary winding turns  $N_1$ , (2) Peak magnetic flux in the core  $\phi_m$ , (3) Full-load primary current when secondary delivers 20,A.

#### Step 1 🚀 : Primary turns $N_1$

$$(N_1 / N_2) = (V_1 / V_2) \Rightarrow (N_1 / 60) = \frac{2200}{220} = 10 \Rightarrow N_1 = 600, \text{ urns}$$

#### Step 2 ⚡ : Peak magnetic flux $\phi_m$ via EMF equation

$$E_2 = 4.44 \cdot f \cdot N_2 \cdot \phi_m \Rightarrow 220 = 4.44 \times 50 \times 60 \times \phi_m$$

$$220 = 13320 \times \phi_m \Rightarrow \phi_m = \frac{220}{13320} = 0.0165, \text{Wb} = 16.5, \text{mWb}$$

#### Step 3 🌀 : Primary current $I_1$

$$I_1 = I_2 \cdot \left( \frac{N_2}{N_1} \right) = 20 \times \left( \frac{60}{600} \right) = 2.0, \text{Amperes}$$

🌀 Primary Turns  $N_1 = 600$  | Peak Flux  $\phi_m = 16.5, \text{mWb}$  | Primary Current  $I_1 = 2.0, \text{A}$ .

### University Question (10 Marks):

A 3300 / 230,V, 50,Hz single-phase transformer has a maximum flux density  $B_m = 1.2, \text{Tesla}$  in the core. The induced EMF per turn is 8.0,Volts.

Calculate: (1) Number of primary and secondary turns, (2) Net cross-sectional area of the core  $A_i$ .

#### Step 1 🚀 : Primary and Secondary Turns

$$\text{EMF per turn} = (E_1 / N_1) = (E_2 / N_2) = 8.0, \text{V/turn}$$

$$N_1 = (E_1 / \text{EMF/turn}) = \frac{3300}{8} = 412.5 \Rightarrow 413, \text{ urns}$$

$$N_2 = (E_2 / \text{EMF/turn}) = \frac{230}{8} = 28.75 \Rightarrow 29, \text{ urns}$$

#### Step 2 ⚡ : Peak Magnetic Flux $\phi_m$

$$\text{EMF per turn} = 4.44 \cdot f \cdot \phi_m \Rightarrow 8.0 = 4.44 \times 50 \times \phi_m$$

$$\phi_m = (8.0 / 222) = 0.03604, \text{Weber} = 36.04, \text{mWb}$$

### Step 3 🌀: Core Cross-Sectional Area $A_i$

$$\phi_m = B_m \times A_i \Rightarrow A_i = (\phi_m / B_m) = (0.03604 / 1.2) = 0.03003, \text{m}^2 = 300.3, \text{cm}^2$$

$$\text{🌀 } N_1 = 413, \text{ turns}, N_2 = 29, \text{ turns} \mid \phi_m = 36.04, \text{mWb} \mid \text{Core Area } A_i = 300.3, \text{cm}^2.$$

## 2. Transformer Losses, Efficiency & The OC/SC Tests

### 1. In-Depth Analysis of Transformer Losses & Efficiency

#### A. Core Losses / Iron Losses ( $P_i$  — Invariant with Load):

Occur inside the magnetic core due to cyclic reversal of magnetization:

**Hysteresis Loss ( $P_h$ ):** Energy dissipated aligning molecular magnetic dipoles:

$$P_h = \eta_h \cdot B_m^{1.6} \cdot f \cdot V_{\text{core}} [\text{Steinmetz Equation}]$$

- Minimized by using **Cold-Rolled Grain-Oriented (CRGO) Silicon Steel** with a narrow hysteresis loop.

**Eddy Current Loss ( $P_e$ ):** Joule heating from induced circular eddy currents in the conductive core:

$$P_e = K_e \cdot B_m^2 \cdot f^2 \cdot t^2 \cdot V_{\text{core}}$$

- Minimized by slicing the solid core into **thin insulated laminations** of thickness  $t \approx 0.25 - 0.35, \text{mm}$  varnished on both sides.

#### B. Copper Losses ( $P_{cu}$  — Variable with Load):

Joule heating in the electrical windings:

$$P_{cu} = I_1^2 R_1 + I_2^2 R_2 = I_2^2 R_{eq2} = x^2 P_{cu,FL}$$

where  $x = \frac{I_2}{I_{2,FL}} = \frac{\text{Load kVA}}{\text{Rated kVA}}$  is the fractional loading factor.

## 2. Efficiency Formulation & Maximum Efficiency Derivation

$$\eta = \frac{\text{Output Power}}{\text{Input Power}} = \frac{x \cdot S_{\text{rated}} \cdot \cos \phi_2}{x \cdot S_{\text{rated}} \cdot \cos \phi_2 + P_i + x^2 P_{\text{cu,FL}}}$$

#### Condition for Maximum Efficiency:

Differentiating  $\eta$  with respect to load current  $I_2$  and setting to zero:

$$\frac{d\eta}{dI_2} = 0 \Rightarrow P_i = x^2 P_{\text{cu,FL}} \Leftrightarrow \text{Constant Iron Loss} = \text{Variable Copper Loss}$$

Fractional load at maximum efficiency:

$$x_{\text{max}} = \frac{P_i}{P_{\text{cu,FL}}}$$

Load kVA for maximum efficiency:

$$\text{kVA}_{\text{max}} = \text{Rated kVA} \times \frac{P_i}{P_{\text{cu,FL}}}$$

### 3. Open Circuit (OC) and Short Circuit (SC) Test Formulations

| Test Metric | Open Circuit (OC) Test | Short Circuit (SC) Test |

| :--- | :--- | :--- |

| **Objective** | Determine Core Loss  $P_i$ ,  $R_0$ , and  $X_0$  | Determine Full-Load Copper Loss  $P_{\text{cu}}$ ,  $R_{\text{eq}}$ ,  $X_{\text{eq}}$  |

| **Winding Energized** | Low Voltage (LV) winding | High Voltage (HV) winding |

| **Opposite Winding** | High Voltage left open ( $I_2 = 0$ ) | Low Voltage shorted with thick copper conductor |

| **Excitation Level** | Rated voltage  $V_1$  at rated frequency  $f$  | Small reduced voltage (5 of rated) |

| **Key Formulae** |  $\cos \phi_0 = \frac{W_{\text{oc}}}{V_1 I_0}$  < divclass = "display - math" > < spanclass = "katex - display" > < spanclass = "katex" > < spanclass = "katex - html" aria - hidden = "true" > < spanclass = "katex - base" > < spanclass = "katex - strut" style = "height : 0.8333em; vertical - align : -0.15em;" > < /span > < spanclass = "mord" > < spanclass = "mordmathnormal" style = "margin - right : 0.0785em;" > < I < /span > < spanclass = "msupsub" > < spanclass = "vlist - tvlist - t2" > < spanclass = "vlist - r" > < spanclass = "vlist" style = "height : 0.1514em;" > < spanstyle = "top : -2.55em; margin - left : -0.0785em; margin - right : 0.05em;" > < spanclass = "pstrut" style = "height : 2.7em;" > < /span > < spanclass = "katex - sizingreset - size6size3mtight" > < spanclass = "mordmathnormalmtight" > c < /span > < /span > < /span > < /span > < spanclass = "vlist - s" > < /span >

$$\frac{1}{\cos^2 \theta} \frac{d\theta}{dt} = \frac{1}{\cos^2 \theta} \frac{d}{dt} \left( \frac{1}{\cos \theta} \right) = \frac{1}{\cos^2 \theta} \left( \frac{1}{\cos^2 \theta} \frac{d\theta}{dt} \right) = \frac{1}{\cos^4 \theta} \frac{d\theta}{dt}$$

$$I_m = I_0 \sin \phi_0 \Rightarrow X_0 = \frac{V_m}{I_m} \mid R_{eq} = \frac{W_{sc}}{I_{sc}^2} < \divclass = "display - math" >$$

$$Z < >$$

$$e < q <$$

$$I <$$

$$V <$$

$$X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

### Problem:

A 10, kVA transformer has iron loss  $P_i = 150$ , W and full-load copper loss  $P_{cu} = 250$ , W. Calculate its full-load efficiency at unity power factor ( $\cos \phi = 1.0$ ).

### Step 1 🚀 : Output Power

$$P_{out} = \text{kVA} \times 1000 \times \cos \phi = 10 \times 1000 \times 1.0 = \mathbf{10,000, \text{ Watts}}$$

### Step 2 ⚡ : Total Losses

$$P_{loss} = P_i + P_{cu} = 150 + 250 = \mathbf{400, \text{ Watts}}$$

### Step 3 🎯 : Efficiency Formula

$$\eta = \frac{P_{out}}{P_{out} + P_{loss}} \times 100 = \frac{10000}{10000 + 400} \times 100 = \frac{10000}{10400} \times 100 = \mathbf{96.15\%}$$

🎯 Full-Load Efficiency:  $\eta = \mathbf{96.15\%}$ .

### University Question (8 Marks):

A 25, kVA transformer has iron loss  $P_i = 200$ , W and full-load copper loss  $P_{cu} = 400$ , W. Find:

Efficiency at half-load with 0.8 power factor lagging.

The load kVA at which maximum efficiency occurs.

### Step 1 🚀 : Half-Load ( $x = 0.5$ ) Calculation

Output power:  $P_{out} = (0.5)(25 \times 10^3)(0.8) = 10,000 \text{ W}$

Copper loss at half-load:  $P_{cu}' = x^2 P_{cu} = (0.5)^2 \times 400 = 0.25 \times 400 = 100 \text{ W}$

Constant iron loss:  $P_i = 200 \text{ W}$

Efficiency:

$$\eta_{50\%} = \frac{10000}{10000 + 100 + 200} \times 100 = \frac{10000}{10300} \times 100 = 97.09\%$$

**Step 2 ⚡ : Load kVA for Maximum Efficiency ( $P_{cu} = P_i$ )**

$$kVA_{\max} = \text{Rated kVA} \times \sqrt{(P_i / P_{cu})} = 25 \times \sqrt{\left(\frac{200}{400}\right)} = 25 \times \sqrt{0.5} = 25 \times 0.7071 = 17.68, \text{kVA}$$

🎯 Half-Load Efficiency = 97.09 | Maximum Efficiency occurs at **17.68**, kVA.

**University Question (12 Marks - University Exam):**

The following test results were obtained on a 5,kVA, 200 / 400,V, 50,Hz single-phase transformer:

**OC Test (LV side):** 200,V, 0.7,A, 60,W

**SC Test (HV side):** 22,V, 12.5,A, 120,W

Find the equivalent circuit parameters referred to the LV side.

GIVEN TEST CONFIGURATION: 5kVA 200V/400V (OC TEST ON LV SIDE & SC TEST ON HV SIDE)

OPEN CIRCUIT (OC) TEST (LV SIDE)

Voltmeter  $V_{oc} = 200$  V (Rated)

Ammeter  $I_{oc} = 0.7$  A (No-load)

Wattmeter  $W_{oc} = 60$  W (Iron Loss  $P_i$ )

HV Terminals: Open-Circuited

SHORT CIRCUIT (SC) TEST (HV SIDE)

Voltmeter  $V_{sc} = 22$  V (Reduced)

Ammeter  $I_{sc} = 12.5$  A (Full-load)

Wattmeter  $W_{sc} = 120$  W (Copper Loss  $P_{cu}$ )

LV Terminals: Thick Copper Short

Determine Shunt ( $R_0$ ,  $X_0$ ) and Series ( $R_{eq1}$ ,  $X_{eq1}$ ) Parameters Referred to LV Side

Fig 2.2a: Open-Circuit and Short-Circuit Diagnostic Test Setup for Single-Phase Transformer

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**Step 1 🚀 : OC Test Parameters (Shunt Branch  $R_0$ ,  $X_0$ )**

$$\cos \phi_0 = \frac{W_{oc}}{V_{oc} I_{oc}} = \frac{60}{200 \times 0.7} = \frac{60}{140} = 0.4286$$

$$\sin \phi_0 = \sqrt{1 - (0.4286)^2} = 0.9035$$

Core loss current:  $I_c = I_{oc} \cos \phi_0 = 0.7 \times 0.4286 = 0.30, \text{A}$

Magnetizing current:  $I_m = I_{oc} \sin \phi_0 = 0.7 \times 0.9035 = 0.632, \text{A}$

$$R_0 = \frac{V_{oc}}{I_c} = \frac{200}{0.30} = \mathbf{666.7, \Omega}$$

$$X_0 = \frac{V_{oc}}{I_m} = \frac{200}{0.632} = \mathbf{316.5, \Omega}$$

**Step 2 ⚡ : SC Test Parameters (Series Branch referred to HV side)**

$$R_{eq2} = \frac{W_{sc}}{I_{sc}^2} = \frac{120}{(12.5)^2} = \frac{120}{156.25} = \mathbf{0.768, \Omega}$$

$$Z_{eq2} = \frac{V_{sc}}{I_{sc}} = \frac{22}{12.5} = \mathbf{1.76, \Omega}$$

$$X_{eq2} = \sqrt{Z_{eq2}^2 - R_{eq2}^2} = \sqrt{(1.76)^2 - (0.768)^2} = \sqrt{3.098 - 0.590} = \mathbf{1.584, \Omega}$$

**Step 3** 🌀: Refer Series Branch to LV Side ( $K = 400/200 = 2$ )

$$R_{eq1} = \frac{R_{eq2}}{K^2} = \frac{0.768}{4} = \mathbf{0.192, \Omega}$$

$$X_{eq1} = \frac{X_{eq2}}{K^2} = \frac{1.584}{4} = \mathbf{0.396, \Omega}$$

SOLVED SCHEMATIC: COMPLETE EQUIVALENT CIRCUIT REFERRED TO LV PRIMARY SIDE

LV+

LV-

$$R_0 = 666.7 \Omega$$

$$X_0 = 316.5 \Omega$$

$$R_{eq1} = 0.192 \Omega$$

$$X_{eq1} = 0.396 \Omega$$

Sec'

Sec'

Solved Model: Shunt  $R_0 = 666.7 \Omega$ ,  $X_0 = 316.5 \Omega$  | Referred Series:  $R_{eq1} = 0.192 \Omega$ ,  $X_{eq1} = 0.396 \Omega$

Fig 2.2b: Solved Single-Phase Transformer Complete Equivalent Circuit Referred to Primary



LV Referred Parameters:  $R_0 = 666.7, \Omega$ ,  $X_0 = 316.5, \Omega$  |  $R_{eq1} = 0.192, \Omega$ ,  $X_{eq1} = 0.396, \Omega$ .

### 3. DC Motors & Generators (Fleming's Hand Rules)

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#### 1. DC Machines: Construction, Action & Hand Rules

A **Direct Current Machine** is an electromechanical energy converter operating bidirectionally:

**DC Generator:** Mechanical shaft torque → DC electrical power via Faraday's Law.

**DC Motor:** DC electrical power → Mechanical shaft torque via Lorentz Force Law.

#### Structural Components:

**Stator:** Yoke (magnetic frame), Field poles with pole shoes, Shunt/Series Field windings.

**Rotor (Armature):** Laminated cylindrical silicon-steel core, insulated armature conductors in slots.

**Commutator:** Split-ring copper wedge segments insulated with mica, acting as a mechanical rectifier/inverter.

**Brushes:** High-grade carbon blocks riding on the commutator to collect or inject current.

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## 2. Fleming's Hand Rules: Left Hand (Motor) vs Right Hand (Generator)

...

FLEMING'S LEFT-HAND RULE (MOTORS) FLEMING'S RIGHT-HAND RULE (GENERATORS)

Thumb = Force (Motion) Thumb = Motion of Conductor

Forefinger = Magnetic Field (N → S) Forefinger = Magnetic Field (N → S)

Middle Finger = Current Middle Finger = Induced Current

...

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## 3. Mathematical Derivations of DC Machine Equations

#### A. General Induced EMF Equation:

Let:

$P$  = Number of poles,  $\phi$  = Magnetic flux per pole (Wb)

$Z$  = Total number of armature conductors,  $N$  = Rotational speed (rpm)

$A$  = Number of parallel paths ( $A = 2$  for Wave winding;  $A = P$  for Lap winding)

Flux cut by one conductor in 1 revolution:  $d\phi = P\phi$ .

Time for 1 revolution:  $dt = \frac{60}{N}$ , seconds.

EMF induced in one conductor:  $e = \frac{d\phi}{dt} = \frac{P\phi}{60/N} = \frac{P\phi N}{60}$ , Volts.

Since  $Z$  conductors are divided into  $A$  parallel paths, conductors in series per path =  $Z/A$ :

$$E_g = \left( \frac{P\phi N}{60} \right) \cdot \frac{Z}{A} = \frac{P\phi ZN}{60A}$$

#### B. DC Motor Back EMF ( $E_b$ ) and Voltage Balance:

$$V = E_b + I_a R_a \Rightarrow I_a = \frac{V - E_b}{R_a}$$

$$E_b = \frac{P\phi ZN}{60A}$$

#### C. Electromagnetic Torque Equation:

Mechanical gross power developed equals electromagnetic power:

$$P_{\text{mech}} = T_e \cdot \omega = T_e \left( \frac{2\pi N}{60} \right) = E_b \cdot I_a$$

$$T_e \left( \frac{2\pi N}{60} \right) = \left( \frac{P\phi ZN}{60A} \right) I_a$$

$$T_e = \frac{1}{2\pi} \left( \frac{PZ}{A} \right) \phi I_a \approx 0.159 \left( \frac{PZ}{A} \right) \phi I_a [\text{N} \cdot \text{m}]$$

$$T_e \propto \phi \cdot I_a$$

### Problem:

A 220, V DC shunt motor has an armature resistance  $R_a = 0.5, \Omega$ . When running at full load, it draws an armature current of 20, A. Calculate the Back EMF  $E_b$ .

GIVEN SCHEMATIC: 220V DC SHUNT MOTOR WITH ARMATURE RESISTANCE  $R_a = 0.5\Omega$  &  $I_a = 20A$

+220V

GND

$I_a = 20\text{ A}$

$R_a = 0.5\ \Omega$

M

+

-

$E_b = ?\text{ V}$

Given:  $V = 220V$ ,  $R_a = 0.5\ \Omega$ ,  $I_a = 20\text{ A} \Rightarrow$  Find Motor Counter-EMF (Back EMF)  $E_b$

Fig 2.3a: DC Shunt Motor Armature Circuit Model with Series Resistance R<sub>a</sub>

---

**Step 1 🚀 : State Motor Voltage Equation**

$$V = E_b + I_a R_a \Rightarrow E_b = V - I_a R_a$$

**Step 2 ⚡ : Substitute Values**

$$E_b = 220 - (20 \times 0.5) = 220 - 10 = \mathbf{210.0, \text{ Volts}}$$

SOLVED SCHEMATIC: BACK EMF BALANCE (E<sub>b</sub> = 210.0V, R<sub>a</sub> DROP = 10.0V)

+220V

0V

R<sub>a</sub> = 0.5 Ω

ΔV = 10 V

M

$$E_b = 210.0 \text{ Volts}$$

$$E_b = 220V - 10V \checkmark$$

$$\text{Solved: } V = E_b + I_a \cdot R_a \Rightarrow E_b = 220V - (20A \times 0.5\Omega) = 220V - 10V = 210.0 \text{ Volts}$$

Fig 2.3b: Solved DC Motor Back EMF Voltage Balance Diagram

🔧 Back EMF generated:  $E_b = 210.0, V$ .

#### University Question (8 Marks):

A 240, V DC shunt motor has armature resistance  $R_a = 0.4, \Omega$  and field resistance  $R_{sh} = 120, \Omega$ .

What is the armature current at the instant of starting (when  $N = 0$ )?

What is the total line current when running at rated speed with  $E_b = 228, V$ ?

---

#### Step 1 🚀 : Starting Armature Current ( $E_b = 0$ )

$$I_{a, \text{start}} = (V - 0 / R_a) = (240 / 0.4) = 600.0, \text{Amperes!}$$

*(Demonstrates why a 3-point starter resistor is mandatory!)*

#### Step 2 ⚡ : Running Armature Current

$$I_a = (V - E_b / R_a) = (240 - 228 / 0.4) = (12 / 0.4) = 30.0, \text{Amperes}$$

#### Step 3 🌀 : Total Line Current $I_L = I_a + I_{sh}$

$$I_{sh} = (V / R_{sh}) = \frac{240}{120} = 2.0, A$$

$$I_L = 30.0 + 2.0 = 32.0, \text{Amperes}$$

🌀 Starting Current = 600.0, A (Dangerous without starter) | Normal Running Line Current = 32.0, A.

#### University Question (10 Marks):

A 230, V DC shunt motor runs at 1000, pm while taking an armature current of 40, A. The armature resistance is  $0.25, \Omega$ .

If the mechanical torque is reduced by 50% and magnetic flux per pole remains constant, find the new operating speed.

---

**Step 1 🚀 : Initial Back EMF  $E_{b1}$** 

$$E_{b1} = V - I_{a1} R_a = 230 - (40 \times 0.25) = 230 - 10 = 220.0, V$$

**Step 2 ⚡ : New Armature Current for 50% Torque**

Since torque  $T \propto \phi I_a$  and flux  $\phi$  is constant:

$$T_2 = 0.5 T_1 \Rightarrow I_{a2} = 0.5 I_{a1} = 0.5 \times 40 = 20.0, A$$

**Step 3 🎯 : New Back EMF  $E_{b2}$** 

$$E_{b2} = V - I_{a2} R_a = 230 - (20 \times 0.25) = 230 - 5 = 225.0, V$$

**Step 4 🎯 : Speed Relation ( $E_b \propto N$ )**

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \Rightarrow N_2 = N_1 \left( \frac{E_{b2}}{E_{b1}} \right) = 1000 \cdot \left( \frac{225}{220} \right) = 1000 \cdot 1.0227 = 1022.7 \text{ rpm}$$

🎯 New Operating Speed:  $N_2 = 1023$ , rpm (Speed rises slightly as load lightens).

## University Solved Examination Questions (Unit 2)

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### MMDU Official University Solved Examination Questions (Unit 2)

**Course:** Basic Electrical & Electronics Engineering ('BELE-001')

**Pattern:** Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (6 Marks In-Depth), D (6 Marks Design/Long)

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EASY

#### Question 4 (1 Mark): Hysteresis vs. Eddy Current Loss Frequency Dependence

##### Question:

How do core losses change if a 50 Hz transformer is operated on a 60 Hz supply at constant voltage?

##### Question Schematic / State Specification:

...

Question Frequency Variation:  $V/f$  ratio changes flux density  $B_m \propto V/f$ .

...

##### Step-by-Step Solution:

##### Solution Analytical Derivation & Schematic:

...

Solution Loss Scaling Proportions:

$$P_h \propto f \cdot (V/f)^{1.6} = V^{1.6} f^{-0.6} \Rightarrow \text{Decreases!}$$

$$P_e \propto f^2 \cdot (V/f)^2 = V^2 \Rightarrow \text{Remains Constant!}$$

...

At constant voltage, flux density decreases as frequency rises. Hysteresis loss decreases slightly (proportional to  $f^{-0.6}$ ) while eddy current loss remains constant, causing total core losses to decrease.



**Verified Result:** Question 4 step-by-step resolution per MMDU university marking scheme.

EASY

### Question 5 (1 Mark): Significance of Back EMF in DC Motors

#### Question:

Why does a DC motor draw dangerously high current if started without an external starter?

#### Question Schematic / State Specification:

...

Question Starting Circuit: At  $N = 0 \Rightarrow$  Back EMF  $E_b = 0 \Rightarrow$  Armature Current  $I_a = V / R_a$ .

...

#### Step-by-Step Solution:

#### Solution Analytical Derivation & Schematic:

...

Solution Current Surge Formula:

$$I_{a,start} = (V - 0) / R_a = 230 \text{ V} / 0.5 \Omega = 460 \text{ A} \approx 15\text{--}20 \times I_{rated}!$$

...

At standstill ( $N=0$ ), back EMF is zero. Because armature resistance ' $R_a$ ' is extremely small ( $\sim 0.5 \Omega$ ), direct connection to rated voltage produces a destructive current surge that will burn the windings unless limited by a starter.



**Verified Result:** Question 5 step-by-step resolution per MMDU university marking scheme.

EASY

### Question 6 (1 Mark): Slip Condition of Induction Motor at Synchronous Speed

#### Question:

Why can an induction motor never run at synchronous speed ( $N = N_s$ )?

#### Question Schematic / State Specification:

...

Question Relative Velocity: Rotor speed  $N$  vs Stator RMF speed  $N_s \Rightarrow$  Relative motion  $\Delta N = N_s - N$ .

...

#### Step-by-Step Solution:

#### Solution Analytical Derivation & Schematic:

...

Solution Induced EMF Zero Vector:

`If  $N = N_s \Rightarrow$  Relative Speed = 0  $\Rightarrow d\Phi/dt = 0 \Rightarrow E_r = 0 \Rightarrow I_r = 0 \Rightarrow T = 0$ !

...

If the rotor reached synchronous speed, rotor conductors would not cut any rotating magnetic flux. Induced rotor EMF, current, and electromagnetic torque would drop to zero, causing the rotor to immediately decelerate.



**Verified Result:** Question 6 step-by-step resolution per MMDU university marking scheme.

**MEDIUM**

### Question 14 (2 Marks): Open-Circuit and Short-Circuit Test Diagnostic Roles

#### Question:

Explain why the wattmeter reading in an Open-Circuit (OC) test represents core loss only, while in a Short-Circuit (SC) test it represents copper loss only.

#### Question Schematic / State Specification:

...

Question Diagnostic Regimes: OC Test [Rated Voltage, Negligible No-Load Current  $I_0$ ] vs SC Test [Rated Current, Tiny Voltage  $V_{sc}$ ].

...

#### Step-by-Step Solution:

#### Solution Analytical Derivation & Schematic:

...

Solution Decoupled Loss Models:

`OC Test:  $I_0 \approx 2-5\% \times I_{rated} \Rightarrow I_0^2 R_{cu} \approx 0 \Rightarrow W_0 = P_{iron}$ `

`SC Test:  $V_{sc} \approx 5-10\% \times V_{rated} \Rightarrow B_m \approx 0 \Rightarrow P_{iron} \approx 0 \Rightarrow W_{sc} = P_{cu}$ `

...

In the OC test, current is so low that copper loss is negligible. In the SC test, the applied voltage is tiny, so core flux and core loss are negligible, isolating the respective loss mechanisms perfectly.



**Verified Result:** Question 14 step-by-step resolution per MMDU university marking scheme.

**HARD****Question 18 (4 Marks): Transformer Efficiency and All-Day Efficiency****Question:**

A 50 kVA distribution transformer has iron loss  $P_i = 500\text{W}$  and full-load copper loss  $P_{cu} = 800\text{W}$ . Calculate: (1) Full-load efficiency at 0.8 pf lagging, (2) Load kVA at which maximum efficiency occurs.

 **Question Schematic / State Specification:**

...

Question Loss Energy Flow: Input Power  $\rightarrow$  [Iron Loss 500W + Copper Loss 800W]  $\rightarrow$  Output Power = kVA  $\times$  pf.

...

**Step-by-Step Solution:** **Solution Analytical Derivation & Schematic:**

...

Solution Efficiency Formulations:

$$1. \text{ Full-Load Output} = 50 \text{ kVA} \times 0.8 = 40 \text{ kW} = 40,000 \text{ W}$$

$$\text{Total Full-Load Losses} = P_i + P_{cu} = 500 + 800 = 1300 \text{ W}$$

$$\text{Efficiency } \eta_{(fl)} = [ 40000 / (40000 + 1300) ] \times 100 = \frac{40000}{41300} \times 100 = 96.85\%$$

$$2. \text{ Load fraction for Maximum Efficiency: } x = \sqrt{P_i / P_{cu}} = \sqrt{\frac{500}{800}} = \sqrt{0.625} = 0.7906$$

$$\text{Load kVA for Max Efficiency} = x \times 50 \text{ kVA} = 0.7906 \times 50 = 39.53 \text{ kVA}$$

...

Maximum efficiency occurs when the transformer is loaded to **39.53 kVA** (79% of rated capacity), where variable copper loss equals 500W.



**Verified Result:** Question 18 step-by-step resolution per MMDU university marking scheme.

**HARD****Question 22 (6 Marks): Complete Theory and Speed Control of DC Shunt Motor****Question:**

Derive the Torque equation of a DC Motor. Describe the two methods of speed control for a DC Shunt Motor (Armature Resistance Control and Field Flux Control) with relevant circuit diagrams, equations, and speed-torque characteristics.

**Question Schematic / State Specification:**

...

Question Speed-Torque Characteristic Slopes: Rated speed  $N_{rated}$   $\rightarrow$  Armature control drops speed below  $N_{rated}$ ; Field control raises speed above  $N_{rated}$ .

...

**Step-by-Step Solution:****Solution Analytical Derivation & Schematic:**

...

Solution Torque & Speed Derivations:

1. Electrical Power converted to Mechanical:  $P_m = E_b I_a = T \cdot \omega$

Substitute  $E_b = \frac{\Phi Z N P}{60 A}$  and  $\omega = 2\pi N / 60$ :

$T \cdot \frac{2\pi N}{60} = \left[ \frac{\Phi Z N P}{60 A} \right] \cdot I_a \Rightarrow T = \left[ \frac{P Z}{2\pi A} \right] \cdot \Phi I_a = 0.159 \Phi Z I_a \text{ (P/A) N-m}$

2. Speed Governing Relation:  $N = \frac{V - I_a R_a}{k \Phi}$

Method A (Armature Resistance):  $R_{ext}$  added in series with  $R_a \Rightarrow N \downarrow$  below rated speed. Wasteful  $I^2 R$  loss.

Method B (Field Flux):  $R_f$  added in series with shunt field  $\Rightarrow I_f \downarrow \Rightarrow \Phi \downarrow \Rightarrow N \uparrow$  above rated speed. Highly efficient!

...

Field control provides a constant-horsepower drive above base speed, whereas armature control provides a constant-torque drive below base speed.



**Verified Result:** Question 22 step-by-step resolution per MMDU university marking scheme.



# Semiconductors, Diodes & Transistors (BJT Made Simple)

**Exam Weightage:** 24 - 28 Marks • **Estimated Study Time:** 35 mins

*Semiconductor band theory simplified; Doping (N-type vs P-type); The PN junction one-way turnstile; Zener voltage regulator design; Rectifier circuits comparison; Clippers & Clampers; BJT transistor explained with water valves.*

## 1. Semiconductors & Doping (The Secret of Silicon)

### 1. Energy Band Theory & Semiconductor Physics

Solids are classified by the width of the forbidden energy bandgap ( $E_g$ ) separating the completely filled **Valence Band** from the empty **Conduction Band**:

**Metals (Conductors):** Valence and conduction bands overlap ( $E_g = 0$ , eV). Massive sea of free electrons.

**Insulators (Diamond, Quartz):** Very wide bandgap ( $E_g > 5$ , eV). Extremely low thermal excitation across gap.

**Semiconductors (Silicon, Germanium):** Moderate bandgap:

$$E_g(\text{Si}) \approx 1.12, \text{eV}, E_g(\text{Ge}) \approx 0.67, \text{eV (at 300, K)}$$

### 2. Intrinsic vs Extrinsic Semiconductors & Doping

**Intrinsic Semiconductor:** Chemically ultra-pure silicon (99.9999999).

- Thermal excitation generates electron-hole pairs:  $n = p = n_i$ .
- At room temperature (300, K),  $n_i \approx 1.5 \times 10^{10}, \text{cm}^{-3}$  for Silicon.

**Extrinsic (Doped) Semiconductors:**

- **N-Type Semiconductor:** Pure silicon doped with Pentavalent impurity atoms (Phosphorus, Arsenic, Antimony). 5 valence electrons  $\rightarrow$  4 form covalent bonds, 1 is easily freed into conduction band. Majority carriers = **Electrons** ( $n \approx N_D$ ), Minority carriers = Holes.
- **P-Type Semiconductor:** Pure silicon doped with Trivalent impurity atoms (Boron, Gallium, Indium). 3 valence electrons  $\rightarrow$  creates a vacant bond state (**Hole**). Majority carriers = **Holes** ( $p \approx N_A$ ), Minority carriers = Electrons.

### 3. The Law of Mass Action & Fermi-Dirac Statistics

● **Mass Action Law:** In any non-degenerate semiconductor at thermal equilibrium, the product of electron and hole concentrations is strictly constant, depending only on temperature and bandgap, regardless of doping:

$$n \cdot p = n_i^2$$

#### Fermi Energy Level ( $E_F$ ):

The energy level at which the Fermi-Dirac probability of electron occupancy is exactly 50 ( $f(E_F) = 0.5$ ):

**Intrinsic:**  $E_F \approx E_i$  lies virtually at the exact center of the bandgap:  $E_i = \frac{E_C + E_V}{2}$ .

**N-Type:** Doping with donors pushes  $E_F$  upward toward the conduction band:

$$E_F - E_i = k_B T \ln\left(\frac{N_D}{n_i}\right)$$

**P-Type:** Doping with acceptors pulls  $E_F$  downward toward the valence band:

$$E_i - E_F = k_B T \ln\left(\frac{N_A}{n_i}\right)$$

#### Problem:

In a pure silicon semiconductor at 300, K, the intrinsic carrier concentration is  $n_i = 1.5 \times 10^{10}, \text{cm}^{-3}$ . The sample is doped with donor atoms to a concentration of  $N_D = 10^{16}, \text{cm}^{-3}$ . Find the majority electron concentration  $n$  and minority hole concentration  $p$ .

#### Step 1 🚀 : Majority Carrier Concentration

Since  $N_D \gg n_i$ , virtually all donor atoms are ionized:

$$n \approx N_D = 1.0 \times 10^{16}, \text{cm}^{-3}$$

## Step 2 ⚡ : Apply Mass Action Law $n \cdot p = n_i^2$

$$p = \frac{n_i^2}{n} = \frac{(1.5 \times 10^{10})^2}{10^{16}} = \frac{2.25 \times 10^{20}}{10^{16}} = 2.25 \times 10^4, \text{cm}^{-3}$$

🔗 Electrons  $n = 10^{16}, \text{cm}^{-3}$  | Holes  $p = 2.25 \times 10^4, \text{cm}^{-3}$  (Drastic suppression of minority carriers).

### University Question (8 Marks):

Pure silicon at room temperature has  $n_i = 1.5 \times 10^{16}, \text{m}^{-3}$ , electron mobility  $\mu_n = 0.135, \text{m}^2/\text{V}\cdot\text{s}$ , and hole mobility  $\mu_p = 0.048, \text{m}^2/\text{V}\cdot\text{s}$ .

Find intrinsic conductivity  $\sigma_i$ .

Find conductivity after doping with  $10^{22}$  donor atoms/ $\text{m}^3$ .

### Step 1 🚀 : Intrinsic Conductivity $\sigma_i = n_i e (\mu_n + \mu_p)$

$$\sigma_i = (1.5 \times 10^{16})(1.6 \times 10^{-19})(0.135 + 0.048) = (2.4 \times 10^{-3})(0.183) = 4.39 \times 10^{-4}, \text{S/m}$$

### Step 2 ⚡ : Doped N-Type Conductivity $\sigma_n \approx ND e \mu_n$

$$\sigma_n = (10^{22})(1.6 \times 10^{-19})(0.135) = (1600)(0.135) = 216.0, \text{S/m}$$

Notice that doping by a tiny fraction increased electrical conductivity by **nearly 500,000 times!**

🔗 Intrinsic Conductivity  $\sigma_i = 4.39 \times 10^{-4}, \text{S/m}$  | Doped Conductivity  $\sigma_n = 216.0, \text{S/m}$ .

### University Question (10 Marks):

Calculate the displacement of the Fermi energy level  $E_F$  from the intrinsic level  $E_i$  in silicon at  $T = 300, \text{K}$  doped with  $2.0 \times 10^{16}, \text{cm}^{-3}$  phosphorus atoms ( $n_i = 1.5 \times 10^{10}, \text{cm}^{-3}$ ,  $k_B T = 0.0259, \text{eV}$ ).

### Step 1 🚀 : Fermi Level Offset Equation

$$E_F - E_i = k_B T \ln\left(\frac{N_D}{n_i}\right)$$

### Step 2 ⚡ : Compute Concentration Ratio

$$(N_D / n_i) = (2.0 \times 10^{16} / 1.5 \times 10^{10}) = 1.333 \times 10^6$$

### Step 3 🔗 : Evaluate Logarithm and Energy Shift

$$\ln(1.333 \times 10^6) = \ln(1.333) + 6 \ln(10) = 0.2877 + 6(2.3026) = 0.2877 + 13.8155 = 14.103$$

$$E_F - E_i = (0.0259, \text{eV}) \times 14.103 = 0.365, \text{eV}$$

The Fermi level shifts **0.365, eV upward toward the conduction band**.

🔗 Fermi Level Shift:  $E_F - E_i = +0.365, \text{eV}$  (closer to Conduction Band).

## 2. PN Junction Diode & The One-Way Turnstile

## 1. The P-N Junction: Formation, Depletion Region & Electrostatics

When P-type and N-type semiconductors are metallurgically joined in a single crystal:

**Diffusion Current:** High hole concentration in P-region causes holes to diffuse into N-region; electrons diffuse from N-region into P-region.

**Uncovered Ion Cores:** Diffusing electrons and holes recombine, leaving unneutralized donor ions ( $N_D^+$ ) on the N-side and unneutralized acceptor ions ( $N_A^-$ ) on the P-side.

**The Depletion Layer (Space Charge Region):** A thin boundary region ( $W \approx 0.1 - 1, \mu\text{m}$ ) completely devoid of mobile charge carriers.

**Built-In Potential Barrier ( $V_0 / V_{bi}$ ):** The space charge establishes an internal electric field  $E$  pointing from N to P that opposes further diffusion:

$$V_0 = V_T \ln \left( \frac{N_A \cdot N_D}{n_i^2} \right)$$

where  $V_T = \frac{k_B T}{q} \approx 26, \text{mV}$  at 300, K.

- For Silicon:  $V_0 \approx 0.7, \text{Volts}$ . For Germanium:  $V_0 \approx 0.3, \text{Volts}$ .

---

## 2. Biasing Regimes & The Shockley Diode Equation

...

Forward Bias (+ to P, - to N) Reverse Bias (- to P, + to N)

$V_{\text{applied}}$  opposes  $V_0$   $V_{\text{applied}}$  aids  $V_0$

Depletion width shrinks:  $W \rightarrow 0$  Depletion width widens:  $W \gg W_0$

Current surges exponentially: Current drops to tiny reverse saturation:

$I_D = I_s \left[ \exp\left\{\frac{V_D}{\eta V_T}\right\} - 1 \right]$   $I_D = -I_s$  (nano-amperes)

...

#### Shockley Diode Characteristic Equation:

$$I_D = I_s \left( e^{\frac{V_D}{\eta V_T}} - 1 \right)$$

$I_s$ : Reverse saturation current ( $10^{-12}, \text{A}$  for Si,  $10^{-6}, \text{A}$  for Ge).

$\eta$ : Ideality factor ( $\eta = 1$  for Ge,  $\eta \approx 1 - 2$  for Si).

---

### 3. Zener Breakdown vs Avalanche Breakdown & Regulators

When reverse bias voltage reaches critical breakdown threshold  $V_{BR}$ :

**Zener Breakdown** ( $V_{BR} < 6, \text{V}$ ): Occurs in **heavily doped junctions** with ultra-thin depletion width ( $< 10, \text{nm}$ ). High internal field ( $> 10^6, \text{V/cm}$ ) directly pulls electrons out of covalent bonds via **quantum mechanical tunneling**. Has a **negative temperature coefficient**.

**Avalanche Breakdown** ( $V_{BR} > 6, \text{V}$ ): Occurs in **lightly doped junctions** with wide depletion layers. High electric field accelerates minority carriers to huge kinetic energies, knocking free secondary electrons via **impact ionization** in a chain reaction. Has a **positive temperature coefficient**.

#### Zener Voltage Regulator Operation:

When biased in the breakdown region, voltage across the Zener stays fixed at  $V_Z$  regardless of current variations ( $I_{Z,\min} \leq I_Z \leq I_{Z,\max}$ ):

$$R_s = \frac{V_{in} - V_Z}{I_Z + I_L}$$

#### Problem:

A silicon diode with knee voltage  $V_0 = 0.7, \text{V}$  is connected in series with a resistor  $R = 220, \Omega$  across a  $5.0, \text{V}$  DC supply. Find the diode current  $I_D$  and voltage across the resistor  $V_R$ .

#### Step 1 🚀 : Check Diode State

Input supply  $5.0, \text{V} > 0.7, \text{V}$ , so the diode is **Forward Biased**.

#### Step 2 ⚡ : Resistor Voltage Drop

$$V_R = V_s - V_D = 5.0 - 0.7 = \mathbf{4.3, \text{Volts}}$$

#### Step 3 🎯 : Diode Current

$$I_D = \frac{V_R}{R} = \frac{4.3, \text{V}}{220, \Omega} = \mathbf{0.0195, \text{A} = 19.55, \text{mA}}$$

🎯 Resistor Voltage  $V_R = 4.3, \text{V}$  | Forward Current  $I_D = 19.55, \text{mA}$ .

#### University Question (8 Marks):

A  $10, \text{V}$  Zener diode is used to regulate a supply voltage that varies from  $15, \text{V}$  to  $20, \text{V}$ . The load resistance is  $R_L = 1000, \Omega$  and minimum Zener current is  $I_{Z,\min} = 5, \text{mA}$ . Calculate the maximum value of series resistor  $R_s$ .

GIVEN SCHEMATIC: ZENER VOLTAGE REGULATOR ( $V_{in} = 15V-20V$ ,  $V_z = 10V$ ,  $R_L = 1000\Omega$ )

$V_{in}$   
(15V-20V)

GND

$R_s = ? \Omega$

$V_Z = 10V$   
 $I_{z,min} = 5mA$

$$R_L = 1000 \, \Omega$$

Given:  $V_{in(min)}=15V$ ,  $V_Z=10V$ ,  $R_L=1000\Omega$ ,  $I_{Z(min)}=5mA \Rightarrow$  Find Maximum Safe Series Resistor  $R_s$

Fig 3.2a: Zener Diode Shunt Voltage Regulator Circuit Topology

---

**Step 1** 🚀 : Load Current  $I_L$

$$I_L = \frac{V_Z}{R_L} = \frac{10, V}{1000, \Omega} = 10.0, mA$$

**Step 2** ⚡ : Minimum Input Voltage Condition ( $V_{in,min} = 15, V$ )

To guarantee regulation at minimum input, the Zener must still carry at least  $I_{Z,min} = 5.0, mA$ .

Total current:  $I_s = I_{Z,min} + I_L = 5.0 + 10.0 = 15.0, mA$ .

**Step 3** 🎯 : Determine Maximum Series Resistor  $R_s$

$$R_{s,max} = \frac{V_{in,min} - V_Z}{I_s} = \frac{15-10}{15 \times 10^{-3}} = \frac{5}{0.015} = 333.3, \Omega$$

SOLVED SCHEMATIC: OPTIMAL CURRENT DISTRIBUTION ( $R_{s,max} = 333.3\Omega$ ,  $I_L = 10mA$ ,  $I_z = 5mA$ )

15V

$$R_s = 333.3 \, \Omega$$

$$I_s = 15 \, \text{mA} \rightarrow$$

$$I_Z = 5.0 \, \text{mA} \downarrow$$

$$I_L = 10 \, \text{mA} \downarrow$$

$$V_{\text{out}} = 10.0 \, \text{V}$$

$$\text{Solved: } I_L = 10\text{V}/1\text{k}\Omega = 10\text{mA} \mid I_s = 5\text{mA} + 10\text{mA} = 15\text{mA} \mid R_s = (15\text{V} - 10\text{V}) / 15\text{mA} = 333.3 \, \Omega$$

Fig 3.2b: Solved Zener Voltage Regulator Circuit Operating at Boundary Condition

 Maximum Series Resistor  $R_{s,\text{max}} = 333.3, \Omega$ .

**University Question (10 Marks):**

A silicon diode has reverse saturation current  $I_s = 10^{-12}\text{A}$  and ideality factor  $\eta = 1$ .

Calculate the diode current at a forward bias of  $V_D = 0.65, V$  at  $T = 300, K$  ( $V_T = 26, mV$ ).

Determine the dynamic (AC) resistance  $r_d$  at this operating point.

**Step 1 🚀 : Apply Shockley Equation**

$$I_D = I_s (e^{V_D/(eta V_T)} - 1)$$

$$\frac{V_D}{V_T} = \frac{0.65}{0.026} = 25.0$$

$$I_D = 10^{-12} (e^{25} - 1) = 10^{-12} (7.20 \times 10^{10}) = 0.0720 \text{ A} = 72.0 \text{ mA}$$

**Step 2 ⚡ : Calculate Dynamic Resistance  $r_d$**

$r_d = (\eta V_T / I_D) = (0.026, V / 0.072, A) = 0.361, \Omega$

🔗 Forward Current  $I_D = 72.0, \text{ mA}$  | Dynamic Resistance  $r_d = 0.361, \Omega$ .

**3. Rectifiers, Clippers & Clampers (AC to Clean DC)**

**1. Rectification Circuits & Performance Analysis**

A **rectifier** converts bidirectional alternating voltage (AC) into pulsating unidirectional direct voltage (DC).

#### Performance Metrics Catalog:

**Rectification Efficiency ( $\eta$ ):**  $\eta = \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (R_f + R_L)}$ .

**Ripple Factor ( $\gamma$ ):** Ratio of RMS AC component to DC component:

$$\gamma = \frac{V_{ac,rms}}{V_{dc}} = \left( \frac{V_{rms}}{V_{dc}} \right)^2 - 1$$

**Peak Inverse Voltage (PIV):** Maximum reverse voltage across a non-conducting diode.

---

## 2. Mathematical Comparison of the Three Standard Rectifier Topologies

| Metric | Half-Wave Rectifier | Center-Tapped Full-Wave | Full-Wave Bridge Rectifier |  
| :--- | :--- | :--- | :--- |  
| **Diode Count**  $\lfloor \frac{1}{2} \rfloor$  | **4** |  
| **Center-Tap Transformer Needed?** | No | **Yes (Expensive, Bulky)** | **No (Standard transformer)** |  
| **Peak Inverse Voltage (PIV)** |  $V_m$  |  $2V_m$  |  $V_m$  (*Economical!*) |  
| **DC Output Voltage ( $V_{dc}$ )** |  $\frac{V_m}{\pi} \approx 0.318V_m$  |  $\frac{2V_m}{\pi} \approx 0.637V_m$  |  $\frac{2V_m}{\pi} \approx \mathbf{0.637V_m}$  |  
| **RMS Output Voltage ( $V_{rms}$ )** |  $\frac{V_m}{2} = 0.50V_m$  |  $\frac{V_m}{2} \approx 0.707V_m$  |  $\frac{V_m}{2} \approx \mathbf{0.707V_m}$  |  
| **Max Efficiency ( $\eta_{max}$ )** | **40.6%** | **81.2%** | **81.2%** |  
| **Ripple Factor ( $\gamma$ )** | **1.21** | **0.482** | **0.482** |  
| **Fundamental Ripple Frequency** |  $f$  (50, Hz) |  $2f$  (100, Hz) |  **$2f$**  (100, Hz, easier to filter!) |

---

## 3. Waveform Shapers: Clippers, Clampers & Filters

**Clippers (Limiters):** Non-linear wave-shaping networks that remove a portion of an input signal above or below a specified reference voltage without distorting the remaining waveform.

**Clampers (DC Restorers):** Networks comprising a diode, resistor, and capacitor that clamp either the positive or negative peak of an input waveform to a specified DC reference level, preserving the original peak-to-peak amplitude  $2V_m$ .

**Capacitor Filter (C-Filter):** A large capacitor placed in parallel with load resistor  $R_L$  charges to peak  $V_m$  and slowly discharges through  $R_L$  with time constant  $\tau = R_L C \gg T$ :

$$V_{r(p-p)} = \frac{I_{dc}}{2fC}, \gamma = \frac{1}{4 \cdot 3fCR_L}$$

---

### Problem:

A full-wave bridge rectifier is fed from a 230, V/12, V transformer. Assuming ideal diodes, calculate the DC output voltage  $V_{dc}$  and Peak Inverse Voltage (PIV) across each diode.

GIVEN SCHEMATIC: FULL-WAVE BRIDGE RECTIFIER FED BY 230V/12V TRANSFORMER

12 V<sub>rms</sub>

4-Diode Bridge

D<sub>1</sub>, D<sub>2</sub>, D<sub>3</sub>, D<sub>4</sub>

PIV = ? V

+DC

−DC

V<sub>dc</sub> = ? V

Given: 12V Secondary RMS supply, Full-wave bridge  $\Rightarrow$  Find Peak  $V_m$ , Average DC  $V_{dc}$ , and Diode PIV

Fig 3.3a: Full-Wave Bridge Rectifier Circuit Schematic

---

**Step 1 🚀 : Secondary Peak Voltage  $V_m$**

$$V_{rms} = 12.0, \text{ V} \Rightarrow V_m = \sqrt{2} \times 12.0 = \mathbf{16.97, \text{ Volts}}$$

**Step 2 ⚡ : DC Output Voltage  $V_{dc}$**

$$V_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 16.97}{\pi} = \mathbf{10.80, \text{ Volts}}$$

**Step 3 🎯 : Peak Inverse Voltage (PIV)**

For a bridge rectifier:

$$\text{PIV} = V_m = \mathbf{16.97, \text{ Volts}}$$

SOLVED SCHEMATIC: DC RECTIFICATION ( $V_m = 16.97\text{V}$ ,  $V_{dc} = 10.80\text{V}$ ,  $\text{PIV} = 16.97\text{V}$ )

Input AC Wave

$$V_{rms} = 12.0 \text{ V}$$

$$V_m = 16.97 \text{ V}$$

$$V_m = \sqrt{2} \times 12V$$

Bridge Action

$$PIV = V_m$$

$$PIV = 16.97 \text{ V}$$

Full-Wave Inversion

Pulsating DC

$$V_{dc} = 10.80 \text{ V}$$

$$V_{dc} = 2 \cdot V_m / \pi$$

$$\text{Ripple freq} = 100\text{Hz}$$

$$\text{Solved: } V_m = \sqrt{2} \times 12V = 16.97 \text{ V} \mid V_{dc} = (2 \times 16.97) / \pi = 10.80 \text{ V} \mid PIV = 16.97 \text{ Volts}$$

Fig 3.3b: Solved Full-Wave Rectifier Signal Flow & Key Metric Specifications

$$\text{🎯 } V_{dc} = 10.80, \text{ V} \mid \text{Diode PIV rating needed} = 16.97, \text{ V.}$$

### University Question (8 Marks):

A full-wave rectifier delivers power to a  $500, \Omega$  load resistor from a secondary voltage of  $v(t) = 50 \sin(314t)$ . Each diode has forward resistance  $R_f = 20, \Omega$ .

Calculate: (1) DC load current  $I_{dc}$ , (2) Rectification efficiency  $\eta$ , (3) Ripple factor  $\gamma$ .

#### Step 1 🚀 : DC Load Current

$$I_m = (V_m / R_f + R_L) = \frac{50}{20 + 500} = \frac{50}{520} = 0.09615, \text{ A}$$

$$I_{dc} = (2 I_m / \pi) = (2 \times 0.09615 / \pi) = 0.0612, \text{ A} = 61.2, \text{ mA}$$

#### Step 2 ⚡ : Rectification Efficiency $\eta$

$$\eta = (P_{dc} / P_{ac}) = (0.812 / 1 + (R_f / R_L)) = (0.812 / 1 + \frac{20}{500}) = (0.812 / 1.04) = 78.08\%$$

#### Step 3 🎯 : Ripple Factor

$$\gamma = \left( \frac{I_{rms}}{I_{dc}} \right)^2 - 1 = (1.11)^2 - 1 = 0.482$$

🌀  $I_{dc} = 61.2, \text{mA}$  | Efficiency  $\eta = 78.08$  | Ripple Factor  $\gamma = 0.482$ .

### University Question (10 Marks):

A full-wave bridge rectifier is supplied by a 50,Hz AC source with peak voltage  $V_m = 24, \text{V}$ . The DC load draws  $I_{dc} = 200, \text{mA}$ .

Design a shunt capacitor filter such that the peak-to-peak ripple voltage  $V_r$  does not exceed  $1.0, \text{V}$ . Also determine the resulting ripple factor  $\gamma$ .

#### Step 1 🚀 : Capacitor Filter Sizing Formula

For full-wave rectifier, ripple frequency is  $2f = 2 \times 50 = 100, \text{Hz}$ .

$$V_{r(p-p)} = \frac{I_{dc}}{2 f C}$$

$$1.0 = (0.20 / 2(50) C) = (0.20 / 100 C) \Rightarrow C = (0.20 / 100) = 2.0 \times 10^{-3}, \text{F} = 2000, \mu\text{F}$$

#### Step 2 ⚡ : DC Output Voltage with Filter

$$V_{dc} = V_m - (V_{r(p-p)} / 2) = 24 - (1.0 / 2) = 23.5, \text{Volts}$$

#### Step 3 🌀 : Ripple Factor $\gamma$

$$\gamma = (V_{r(rms)} / V_{dc}) = (V_{r(p-p)} / 2\sqrt{3}) \cdot V_{dc} = (1.0 / 2\sqrt{3}) \times 23.5 = (1.0 / 81.41) = 0.0123 = 1.23\%$$

🌀 Filter Capacitor  $C = 2000, \mu\text{F}$  | Output  $V_{dc} = 23.5, \text{V}$  | Ripple Factor  $\gamma = 1.23$  (Ultra-smooth DC).

## 4. Bipolar Junction Transistor (BJT) — The Valve Analogy

### 1. Bipolar Junction Transistors (BJT): Physical Structure & Working

A **Bipolar Junction Transistor** is a three-terminal, two-junction semiconductor device whose operation relies on transport of both majority and minority charge carriers (bipolar).

#### The Three Structural Regions:

**Emitter (E):** Heavily doped to inject maximum majority carriers into the base. Moderate physical size.

**Base (B):** Ultra-thin ( $< 1, \mu\text{m}$ ) and extremely lightly doped to minimize electron-hole recombination ( $< 2$ ).

**Collector (C):** Moderately doped, physically the largest region to safely dissipate Joule heat generated by collecting carriers.

### 2. The Three Operating Regimes & Biasing Configurations

| Operating Region | Emitter-Base Junction (EBJ) | Collector-Base Junction (CBJ) | Circuit Application |

| :--- | :--- | :--- | :--- |

| **Active Region** | **Forward Biased** | **Reverse Biased** | Linear Analog Amplifier |  
| **Saturation Region** | **Forward Biased** | **Forward Biased** | Digital Switch ON (Closed) ( $V_{CE,sat} \approx 0.2, V$ ) |  
| **Cutoff Region** | **Reverse Biased** | **Reverse Biased** | Digital Switch OFF (Open) ( $I_C \approx 0$ ) |  
| **Inverse Active** | **Reverse Biased** | **Forward Biased** | Attenuator (Rarely used) |

---

### 3. Current Gain Relations ( $\alpha, \beta, \gamma$ ) & Characteristics

**Current Continuity Equation:**

$$I_E = I_B + I_C$$

**Common-Base Current Gain ( $\alpha$ ):**

$$\alpha = \frac{I_C}{I_E} \approx 0.95 - 0.998$$

**Common-Emitter Current Gain ( $\beta$  or  $h_{FE}$ ):**

$$\beta = \frac{I_C}{I_B} \approx 50 - 300$$

**Mathematical Derivation of  $\beta$  in terms of  $\alpha$ :**

$$I_E = I_B + I_C \Rightarrow \frac{I_E}{I_C} = \frac{I_B}{I_C} + 1 \Rightarrow \frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\frac{1}{\beta} = \frac{1}{\alpha} - 1 = \frac{1 - \alpha}{\alpha} \Rightarrow \beta = \frac{\alpha}{1 - \alpha}, \alpha = \frac{\beta}{\beta + 1}$$

#### DC Operating Point (Q-Point) and Load Line:

Applying KVL to the collector-emitter loop of a Common-Emitter amplifier:

$$V_{CE} = V_{CC} - I_C R_C$$

Upper intercept (Saturation limit):  $V_{CE} = 0 \Rightarrow I_{C,\text{sat}} = \frac{V_{CC}}{R_C}$ .

Lower intercept (Cutoff limit):  $I_C = 0 \Rightarrow V_{CE,\text{cutoff}} = V_{CC}$ .

### Problem:

A transistor has a base current  $I_B = 50, \mu\text{A}$  and collector current  $I_C = 4.95, \text{mA}$ . Find: (1) Emitter current  $I_E$ , (2) Current gain  $\beta$ , (3) Current gain  $\alpha$ .

**Step 1 🚀 : Emitter Current**  $I_E = I_C + I_B$

$$I_E = 4.95, \text{mA} + 0.050, \text{mA} = \mathbf{5.0, \text{mA}}$$

**Step 2 ⚡ : Current Gain  $\beta$**

$$\beta = \frac{I_C}{I_B} = \frac{4.95 \times 10^{-3}}{50 \times 10^{-6}} = \mathbf{99.0}$$

**Step 3 🎯 : Current Gain  $\alpha$**

$$\alpha = \frac{\beta}{\beta + 1} = \frac{99}{100} = \mathbf{0.99}$$

🎯  $I_E = 5.0, \text{mA}$  | Common-Emitter Gain  $\beta = 99$  | Common-Base Gain  $\alpha = 0.99$ .

### University Question (8 Marks):

In a Common-Emitter BJT circuit,  $V_{CC} = 12, \text{V}$ ,  $R_C = 2.0, \text{k}\Omega$ ,  $R_B = 200, \text{k}\Omega$ , and  $\beta = 100$ . Assuming  $V_{BE} = 0.7, \text{V}$ , find:

Base current  $I_B$ .

Operating point ( $I_C$ ,  $V_{CE}$ ).

GIVEN SCHEMATIC: COMMON-EMITTER BJT STAGE ( $V_{CC} = 12V$ ,  $R_C = 2k\Omega$ ,  $R_B = 200k\Omega$ ,  $\beta = 100$ )

+12V ( $V_{CC}$ )

$R_B = 200\text{ k}\Omega$

$I_B \rightarrow$

$R_C = 2.0\text{ k}\Omega$

C

BJT Parameters

$$\beta = 100$$

$$V_{BE} = 0.70 \text{ V}$$

NPN Silicon BJT

Given:  $V_{CC} = 12\text{V}$ ,  $R_C = 2\text{k}\Omega$ ,  $R_B = 200\text{k}\Omega$ ,  $\beta = 100 \Rightarrow$  Find Base Current  $I_B$  and Q-Point ( $I_C$ ,  $V_{CE}$ )

Fig 3.4a: Common-Emitter Fixed-Base Bias Circuit Schematic

---

**Step 1 🚀 : Base Current  $I_B$**

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 - 0.7}{200 \times 10^3} = \frac{11.3}{200,000} = 56.5, \mu\text{A}$$

**Step 2 ⚡ : Collector Current  $I_C$**

$$I_C = \beta I_B = 100 \times (56.5 \times 10^{-6}) = 5.65, \text{mA}$$

**Step 3 🎯 : Collector-Emitter Voltage  $V_{CE}$**

$$V_{CE} = V_{CC} - I_C R_C = 12 - (5.65 \times 10^{-3})(2000) = 12 - 11.3 = 0.70, \text{Volts}$$

The transistor is in the active region near the edge of saturation!

SOLVED SCHEMATIC: BIAS OPERATING POINT ( $I_B = 56.5\mu A$ ,  $I_C = 5.65mA$ ,  $V_{CE} = 0.70V$ )

DC LOAD LINE & Q-POINT

$V_{CE}$

$I_C$

6mA

12V

Q-Point

QUIESCENT OPERATING STATE

1. Base Current:  $I_B = (12 - 0.7)/200k = 56.5 \mu A$

2. Collector Current:  $I_C = \beta \cdot I_B = 5.65 \text{ mA}$

3.  $V_{CE} = 12 - (5.65mA \times 2k\Omega) = 0.70 \text{ Volts}$

Status: In Active Region near saturation threshold

Solved:  $I_B = 56.5 \mu A$  |  $I_C = 5.65 \text{ mA}$  |  $V_{CE} = 0.70 \text{ V}$  | Load Line Q-Point Confirmed

Fig 3.4b: Solved Common-Emitter Load Line and DC Operating Point (Q-Point)

🎯 Q-Point:  $I_C = 5.65, \text{mA}$ ;  $V_{CE} = 0.70, \text{V}$ .

#### University Question (12 Marks - University Exam):

Design a voltage-divider biased Common-Emitter amplifier stage with  $V_{CC} = 15, \text{V}$ , quiescent point  $I_C = 2.0, \text{mA}$ ,  $V_{CE} = 7.5, \text{V}$ , and stability factor condition  $V_E = 0.1 V_{CC}$ . Transistor has  $\beta = 100$ ,  $V_{BE} = 0.7, \text{V}$ . Find  $R_E$ ,  $R_C$ ,  $R_1$ ,  $R_2$ .

---

#### Step 1 🚀 : Determine $R_E$ and $R_C$

$$V_E = 0.1 V_{CC} = 0.1(15) = 1.5, \text{V}$$

$$R_E = (V_E / I_E) \approx (V_E / I_C) = (1.5, \text{V} / 2.0, \text{mA}) = 750, \Omega$$

From collector loop:  $V_{CC} = V_{CE} + I_C R_C + V_E$ :

$$15 = 7.5 + (2.0 \times 10^{-3}) R_C + 1.5 \Rightarrow 2.0 \times 10^{-3} R_C = 6.0 \Rightarrow R_C = 3.0, \text{k}\Omega$$

#### Step 2 ⚡ : Base Voltage $V_B$

$$V_B = V_E + V_{BE} = 1.5 + 0.7 = 2.2, \text{Volts}$$

#### Step 3 🎯 : Size Bias Resistors $R_1$ and $R_2$

For excellent thermal stability, make divider current  $I_1 \approx 10 I_B$ :

$$I_B = (I_C / \beta) = (2.0, \text{mA} / 100) = 20, \mu\text{A} \Rightarrow I_{div} = 10 I_B = 200, \mu\text{A}$$

$$R_2 = (V_B / I_{div}) = (2.2, \text{V} / 200, \mu\text{A}) = 11.0, \text{k}\Omega$$

$$R_1 = (V_{CC} - V_B / I_{div}) = (15 - 2.2 / 200, \mu\text{A}) = (12.8 / 200, \mu\text{A}) = 64.0, \text{k}\Omega$$

🎯 Designed Bias Stage:  $R_C = 3.0, \text{k}\Omega$ ,  $R_E = 750, \Omega$ ,  $R_1 = 64, \text{k}\Omega$ ,  $R_2 = 11, \text{k}\Omega$ .

### University Solved Examination Questions (Unit 3)

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#### MMDU Official University Solved Examination Questions (Unit 3)

**Course:** Basic Electrical & Electronics Engineering ('BELE-001')

**Pattern:** Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (6 Marks In-Depth), D (6 Marks Design/Long)

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EASY

### Question 7 (1 Mark): Barrier Potential of Silicon vs. Germanium

#### Question:

Why is the forward knee voltage of Silicon (0.7V) higher than that of Germanium (0.3V)?

#### Question Schematic / State Specification:

...

Question Bandgap Atomic Dimension: Si ( $E_g = 1.1$  eV, tightly bound valence electrons) vs Ge ( $E_g = 0.67$  eV).

...

#### Step-by-Step Solution:

#### Solution Analytical Derivation & Schematic:

...

Solution Barrier Height Equation:

$V_0 = V_T \ln(N_A N_D / n_i^2)$ . Higher bandgap in Si yields lower intrinsic carrier density  $n_i$   $\Rightarrow$  higher built-in potential  $V_0$ .

...

Silicon's larger energy bandgap (1.1 eV vs 0.67 eV) creates a stronger internal electric field across the depletion region, requiring 0.7V of forward bias to overcome compared to 0.3V for Germanium.



**Verified Result:** Question 7 step-by-step resolution per MMDU university marking scheme.

EASY

### Question 8 (1 Mark): Peak Inverse Voltage in Bridge vs. Center-Tapped Rectifier

#### Question:

Compare the Peak Inverse Voltage (PIV) rating demanded of diodes in Bridge vs Center-Tapped rectifiers.

#### Question Schematic / State Specification:

...

Question Reverse Voltage Stress: Bridge Diode sees  $V_m$  | Center-Tapped Diode sees  $2 V_m$ .

...

#### Step-by-Step Solution:

#### Solution Analytical Derivation & Schematic:

...

Solution PIV Ratio Formulation:

$\text{PIV}_{\text{Bridge}} = V_m$  versus  $\text{PIV}_{\text{Center-Tap}} = 2 V_m$ .

...

In a center-tapped full-wave rectifier, each non-conducting diode must withstand the full secondary voltage  $2 V_m$ . In a bridge rectifier, the reverse voltage divides across two reverse diodes, requiring only  $V_m$  rating.



**Verified Result:** Question 8 step-by-step resolution per MMDU university marking scheme.

EASY

### Question 9 (1 Mark): BJT Base Width Design Principle

#### Question:

Why is the base region of a BJT manufactured ultra-thin and lightly doped?

#### Question Schematic / State Specification:

...

Question Carrier Diffusion Path: Emitter (Heavy Injection) → Base (Minimal Recombination) → Collector (Extraction).

...

#### Step-by-Step Solution:

#### Solution Analytical Derivation & Schematic:

...

Solution Recombination Minimization Model:

`Base Recombination < 2%  $\Rightarrow I_C \approx I_E \Rightarrow \alpha = I_C/I_E \approx 0.98-0.99`.$

...

A thin, lightly doped base ensures that almost all injected minority carriers traverse the base without recombining, allowing over 98% of carriers to be swept into the collector for high current gain  $\beta$ .



**Verified Result:** Question 9 step-by-step resolution per MMDU university marking scheme.

**MEDIUM**

### Question 15 (2 Marks): Zener Shunt Regulator Load Current Calculation

#### Question:

A 10V Zener regulator has input  $V_{in} = 20V$ , series resistor  $R_S = 200\Omega$ , and load  $R_L = 1\text{ k}\Omega$ . Calculate source current  $I_S$ , load current  $I_L$ , and Zener current  $I_Z$ .

#### Question Schematic / State Specification:

...

Question Shunt Circuit Diagram: 20V DC  $\rightarrow$   $R_S$  ( $200\Omega$ )  $\rightarrow$  Node  $V_Z$  (10V)  $\rightarrow$  Parallel branches: Zener Diode &  $R_L$  ( $1\text{ k}\Omega$ ).

...

#### Step-by-Step Solution:

#### Solution Analytical Derivation & Schematic:

...

Solution Current Distribution Formulation:

$$I_S = (V_{in} - V_Z) / R_S = (20 - 10) / 200 = 10 / 200 = 0.05\text{ A} = 50\text{ mA}$$

$$I_L = V_Z / R_L = 10\text{ V} / 1000\ \Omega = 0.01\text{ A} = 10\text{ mA}$$

$$I_Z = I_S - I_L = 50\text{ mA} - 10\text{ mA} = 40\text{ mA}$$

...

The Zener diode safely conducts 40 mA, maintaining a steady 10V regulated output across the  $1\text{ k}\Omega$  load.



**Verified Result:** Question 15 step-by-step resolution per MMDU university marking scheme.

**HARD****Question 19 (4 Marks): Derivation of Full-Wave Bridge Rectifier Ripple Factor****Question:**

Derive the mathematical expression for Ripple Factor  $\gamma$  of a Full-Wave Bridge Rectifier and prove it equals 0.482.

**Question Schematic / State Specification:**

...

Question Rectified Waveform: Pulsating DC with period  $T/2$ :  $v(t) = |V_m \sin(\omega t)|$ .

...

**Step-by-Step Solution:****⚡ Solution Analytical Derivation & Schematic:**

...

Solution Ripple Factor Derivation Trace:

$$\text{Ripple Factor } \gamma = V_{ac,rms} / V_{dc} = \sqrt{(V_{rms}/V_{dc})^2 - 1}$$

$$V_{dc} = \frac{1}{\pi} \int_0^\pi V_m \sin(\theta) d\theta = (2 V_m) / \pi \approx 0.6366 V_m$$

$$V_{rms} = \sqrt{\frac{1}{\pi} \int_0^\pi V_m^2 \sin^2(\theta) d\theta} = V_m / \sqrt{2} \approx 0.7071 V_m$$

$$\text{Form Factor } K_f = V_{rms} / V_{dc} = \frac{V_m / \sqrt{2}}{(2 V_m) / \pi} = \pi / (2 \sqrt{2}) \approx 1.1107$$

$$\gamma = \sqrt{(1.1107)^2 - 1} = \sqrt{1.2337 - 1} = \sqrt{0.2337} = 0.4834 \approx 0.482$$

...

The ripple factor evaluates to 0.482, indicating that AC harmonic content is less than half of the DC component without filtering.



**Verified Result:** Question 19 step-by-step resolution per MMDU university marking scheme.

**HARD****Question 23 (6 Marks): BJT Common Emitter Input and Output Characteristics****Question:**

Sketch and thoroughly explain the Input and Output static characteristics of an NPN Bipolar Junction Transistor in Common Emitter (CE) configuration. Define Early Effect (Base-Width Modulation) and explain how the BJT acts as a digital switch.

**Question Schematic / State Specification:**

...

Question Characteristic Plots: Input ( $I_B$  vs  $V_{BE}$  at constant  $V_{CE}$ ) | Output ( $I_C$  vs  $V_{CE}$  at constant  $I_B$  showing Cutoff, Active, and Saturation).

...

**Step-by-Step Solution:****Solution Analytical Derivation & Schematic:**

...

Solution 3 Operating Regimes & Early Effect:

- `1. Cutoff ( $V_{BE} \leq 0$ ,  $V_{CE} = V_{CC}$ ): Both junctions reverse biased;  $I_C \approx 0 \Rightarrow$  Switch OFF (Open)`
- `2. Saturation ( $V_{BE} > 0.7V$ ,  $V_{CE} \approx 0.2V$ ): Both junctions forward biased;  $I_C = I_{C,sat} \Rightarrow$  Switch ON (Closed)`
- `3. Active ( $V_{BE} > 0.7V$ ,  $V_{CE} > 1V$ ): EBJ forward, CBJ reverse; linear relation  $I_C = \beta I_B \Rightarrow$  Linear Amplifier`

`Early Effect: As  $V_{CE} \uparrow \Rightarrow$  Reverse bias on CBJ  $\uparrow \Rightarrow$  Depletion width  $\uparrow \Rightarrow$  Effective base width  $W_b \downarrow \Rightarrow I_C$  slopes upward`

...

The Early Effect causes collector current curves to have a slight positive slope in the active region; projecting these curves backward intercepts the negative voltage axis at the Early Voltage  $-V_A$ .



**Verified Result:** Question 23 step-by-step resolution per MMDU university marking scheme.



# Number Systems & Boolean Logic (How Computers Think)

**Exam Weightage:** 20 - 24 Marks • **Estimated Study Time:** 35 mins

*Counting in Decimal, Binary, Octal, and Hexadecimal; 2's complement subtraction with zero tears; BCD addition with the famous +6 correction; Excess-3 self-complementing magic; Boolean algebra laws; De Morgan's theorems proven with 4 NAND gates.*

## 1. Number Systems & 2's Complement Subtraction

### 1. Positional Number Systems & Inter-Radix Conversions

A positional number system represents numeric values using radix  $r$  (base) and a set of  $r$  discrete symbols  $0, 1, \dots, r-1$ :

$$N = \sum_{i=-m}^{n-1} d_i \cdot r^i = d_{n-1}r^{n-1} + \dots + d_1r^1 + d_0r^0 + d_{-1}r^{-1} + \dots + d_{-m}r^{-m}$$

#### Common Digital Number Radices:

**Decimal** ( $r = 10$ ): Human standard: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

**Binary** ( $r = 2$ ): Digital logic standard: 0, 1.

**Octal** ( $r = 8 = 2^3$ ): Grouping of 3 binary bits: 0, 1, 2, 3, 4, 5, 6, 7.

**Hexadecimal** ( $r = 16 = 2^4$ ): Grouping of 4 binary bits:  
0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A(10), B(11), C(12), D(13), E(14), F(15).

### 2. Signed Number Representations & 2's Complement Arithmetic

Computers represent signed integers using fixed-width registers ( $n$  bits):

**Sign-Magnitude:** MSB is sign (0 = +, 1 = -), remaining  $(n-1)$  bits represent magnitude. Has dual representation of zero (+0 and -0).

**1's Complement:** Negative number obtained by inverting every bit:  $[-A]_{1's} = (2^n - 1) - A$ .

**2's Complement (The Modern Processor Gold Standard):**

$$[-A]_{2's} = 2^n - A = [-A]_{1's} + 1$$

- Unique representation of zero (00000000<sub>2</sub>).
- Asymmetric range for  $n$  bits:  $[-2^{n-1}, +2^{n-1} - 1]$ .

### 3. Rigorous Rules of 2's Complement Subtraction ( $A - B$ )

Compute  $A + [-B]_{2's}$ :

#### Case 1: $A \geq B$ (Positive Result):

- Addition yields an **End Carry of 1**.
- **Rule:** Discard the end carry. The remaining  $n$ -bit sum is strictly positive and directly represents the correct magnitude!

#### Case 2: $A < B$ (Negative Result):

- Addition yields **NO End Carry (0)**.
- **Rule:** The result is strictly negative and stored in 2's complement form. To recover the true decimal magnitude, take the 2's complement of the sum and attach a minus sign!

#### Problem:

Convert decimal number (53)<sub>10</sub> into: (1) Binary, (2) Hexadecimal.

#### Step 1 🚀 : Repeated Division by 2 for Binary

$$53 \div 2 = 26, \text{ remainder } 1$$

$$26 \div 2 = 13, \text{ remainder } 0$$

$$13 \div 2 = 6, \text{ remainder } 1$$

$$6 \div 2 = 3, \text{ remainder } 0$$

$$3 \div 2 = 1, \text{ remainder } 1$$

$$1 \div 2 = 0, \text{ remainder } 1$$

Read remainders bottom-to-top: (53)<sub>10</sub> = (110101)<sub>2</sub>.

#### Step 2 ⚡ : Convert to Hexadecimal (Group in 4s)

$$(0011; 0101)_2 = (3; 5)_{16} = (35)_{16}$$

Check:  $3 \times 16 + 5 = 48 + 5 = 53$ .

🎯  $(53)_{10} = (110101)_2 = (35)_{16}$ .

### University Question (6 Marks):

Perform the subtraction  $(28)_{10} - (13)_{10}$  using 8-bit 2's complement arithmetic.

---

#### Step 1 🚀 : Convert Numbers to 8-Bit Binary

$$+28 = 00011100_2$$

$$+13 = 00001101_2$$

#### Step 2 ⚡ : Find 2's Complement of 13 ( $[-13]_2$ 's)

1's complement of 13:  $11110010_2$

Add 1:  $11110010 + 1 = 11110011_2$

#### Step 3 🎯 : Add +28 and $[-13]_2$ 's

$$\begin{array}{r@{}} 00011100 & (+28) \\ + 11110011 & (-13) \\ \hline (1), 00001111 & (\text{Sum}) \end{array}$$

#### Step 4 🎯 : Handle End Carry

End carry of 1 is generated  $\Rightarrow$  **Discard end carry!**

Result is positive:  $00001111_2 = 8 + 4 + 2 + 1 = +15_{10}$ . Matches  $28 - 13 = 15$ !

🎯 Result:  $00001111_2 = +15_{10}$ . End carry discarded.

### University Question (8 Marks):

Perform the subtraction  $(14)_{10} - (39)_{10}$  using 8-bit 2's complement arithmetic. Explain how the negative result is recognized and evaluated.

---

#### Step 1 🚀 : 8-bit Binary Representations

$$+14 = 00001110_2$$

$$+39 = 00100111_2$$

#### Step 2 ⚡ : 2's Complement of Subtrahend +39

1's complement:  $11011000$

2's complement:  $11011000 + 1 = 11011001_2$

### Step 3 🌟: Add 14 + [-39]₂'s

$\begin{array}{r} 0001110 \\ + 11011001 \\ \hline \end{array}$

0001110 & (+14) \\n+ 11011001 & (-39) \\n\\hline

11100111 & (No End Carry!)

$\end{array}$

### Step 4 🎯: Interpret Result without End Carry

Since there is **NO end carry**, the result is **strictly negative** and stored in 2's complement form.

To find magnitude, take 2's complement of the sum:

1's complement of 11100111: 00011000

Add 1:  $00011000 + 1 = 00011001_2 = 16 + 8 + 1 = 25$

Attach negative sign:  $-25_{10}$ . Matches  $14 - 39 = -25$ !

🌟 No end carry  $\Rightarrow$  Result is Negative:  $-(00011001)_2 = -25_{10}$ .

## 2. BCD Code (+6 Rule) & Excess-3 Self-Complementing Magic

### 1. Binary Coded Decimal (BCD) & The +6 Correction Rule

#### 8421 BCD (Natural BCD) Encoding:

In 8421 BCD, each decimal digit (0 through 9) is encoded as an independent 4-bit nibble using weights 8, 4, 2, 1:

$$0_{10} = 0000_{\text{BCD}}, \dots, 9_{10} = 1001_{\text{BCD}}$$

A 4-bit register has  $2^4 = 16$  possible binary states ( $0000_2$  to  $1111_2$ ).

The 6 code words from  $1010_2$  ( $10_{10}$ ) **through**  $1111_2$  ( $15_{10}$ ) are **ILLEGAL / FORBIDDEN states**!

### 2. Analytical Reason for the +6 Correction in BCD Addition

When two BCD digits are added using binary adders:

**Condition for Correction:** If the 4-bit binary sum exceeds 9 ( $> 1001_2$ ) OR generates an auxiliary carry out of the nibble ( $C_{\text{out}} = 1$ ), the sum has stepped into or through the forbidden zone!

**The +6 Rule:** The difference between hexadecimal radix (16) and decimal radix (10) is  $16 - 10 = 6$ . Adding binary  $0110_2$  (+6) skips directly past the 6 invalid states, forcing the correct decimal carry into the next higher

nibble:

$$\text{BCD Sum} = (\text{Binary Sum}) + 0110_2 (\text{if sum} > 9 \text{ or carry} = 1)$$

### 3. Excess-3 (XS-3) Code & Self-Complementing Property

Excess-3 is an unweighted code obtained by adding binary  $0011_2$  (+3) to each BCD digit:

$$\text{XS-3} = \text{BCD} + 0011_2$$

#### The Self-Complementing Theorem:

● **Theorem:** If an Excess-3 code word is bitwise inverted (taking 1's complement), the resulting code word is identical to the Excess-3 representation of the **9's complement** of the original decimal digit!

$$\overline{\text{XS-3}(d)} = \text{XS-3}(9 - d)$$

**Engineering Advantage:** Allows digital arithmetic units to perform decimal subtraction using standard adders and simple inverter gates without complex borrow-propagation circuitry!

#### Problem:

Convert the decimal number  $(47)_{10}$  into: (1) 8421 BCD code, (2) Excess-3 code.

**Step 1** 🚀 : 8421 BCD (Convert each digit individually into 4 bits)

$$4 = 0100_2$$

$$7 = 0111_2$$

$$(47)_{10} = 0100; 0111_{\text{BCD}}$$

**Step 2** ⚡ : Excess-3 Code (Add 3 to each digit)

For digit 4:  $4 + 3 = 7 = 0111_2$

For digit 7:  $7 + 3 = 10 = 1010_2$

$$(47)_{10} = \mathbf{0111; 1010}_{XS-3}$$

🎯  $(47)_{10} = \mathbf{0100; 0111}_{BCD} = \mathbf{0111; 1010}_{XS-3}$ .

### University Question (6 Marks):

Perform the addition  $(38)_{BCD} + (45)_{BCD}$  using BCD arithmetic.

#### Step 1 🚀 : Binary Addition of Nibbles

$$38 = 0011 ; 1000$$

$$45 = 0100 ; 0101$$

$$\begin{array}{r} 0011 \ \& \ 1000 \\ 0100 \ \& \ 0101 \\ \hline 0111 \ \& \ 1101 \text{ (7 and 13)} \end{array}$$

#### Step 2 ⚡ : Check for Illegal Nibbles ( $> 9$ )

Lower nibble is  $1101_2 = 13 > 9 \Rightarrow$  **INVALID BCD!**

Higher nibble is  $0111_2 = 7 \leq 9 \Rightarrow$  Valid.

#### Step 3 🎯 : Add +6 (0110<sub>2</sub>) to Lower Nibble

$$\begin{array}{r} 0111 \ \& \ 1101 \\ 0000 \ \& \ 0110 \\ \hline 1000 \ \& \ 0011 \text{ (8 and 3)} \end{array}$$

$1000_2 = 8$  and  $0011_2 = 3 \Rightarrow (83)_{10}$ . Matches  $38 + 45 = 83$ !

🎯 BCD Sum:  $\mathbf{1000; 0011}_{BCD} = (\mathbf{83})_{10}$ .

### University Question (8 Marks):

Demonstrate the self-complementing property of Excess-3 code by finding the 9's complement of  $(62)_{10}$  using bitwise complementation in Excess-3.

#### Step 1 🚀 : Represent $(62)_{10}$ in Excess-3

$$6 + 3 = 9 = 1001_2$$

$$2 + 3 = 5 = 0101_2$$

$$(62)_{10} = 1001 ; 0101_{XS-3}$$

### Step 2 ⚡ : Take Bitwise 1's Complement (Flip Every Bit)

$$\overline{1001 ; 0101} = 0110 ; 1010_2$$

### Step 3 🌟 : Decode Back to Decimal by Subtracting 3

$$\text{Higher nibble: } 0110_2 = 6 \Rightarrow 6 - 3 = 3$$

$$\text{Lower nibble: } 1010_2 = 10 \Rightarrow 10 - 3 = 7$$

Decoded number is  $37_{10}$ .

### Step 4 🧩 : Mathematical Verification

The 9's complement of 62 is:

$$99 - 62 = 37_{10}$$

The bitwise NOT gate directly executed decimal 9's complement subtraction with zero arithmetic!

🌟 Proved: Excess-3 bitwise flip yields  $0110; 1010_{XS-3} = 37_{10}$ , exactly the 9's complement of 62.

## 3. Boolean Algebra & De Morgan's Theorems

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### 1. Boolean Algebra Postulates & Fundamental Theorems

Boolean algebra is an algebraic system defined on the set  $B = 0, 1$  with two binary operators: AND ( $\cdot$ ), OR ( $+$ ), and one unary operator: NOT ( $'$  or  $\overline{\phantom{x}}$ ).

#### Huntington's Postulates & Laws:

**Identity Laws:**  $A + 0 = A, A \cdot 1 = A$

**Null / Dominance Laws:**  $A + 1 = 1, A \cdot 0 = 0$

**Idempotent Laws:**  $A + A = A, A \cdot A = A$

**Complementarity Laws:**  $A + \overline{A} = 1, A \cdot \overline{A} = 0$

**Involution Law:**  $\overline{\overline{A}} = A$

**Commutative Laws:**  $A + B = B + A, A \cdot B = B \cdot A$

**Associative Laws:**  $A + (B + C) = (A + B) + C, A(BC) = (AB)C$

**Distributive Laws:**

$$- A \cdot (B + C) = AB + AC$$

$$- A + BC = (A + B)(A + C) \text{ (Dual distributive law — very powerful!)}$$

$$\textbf{Absorption Laws: } A + AB = A, A(A + B) = A, A + \bar{A}B = A + B$$


---

## 2. De Morgan's Dual Laws & Universal Logic Gates

● **First Theorem (NOR Gate Equivalence):** The complement of a logical sum is equal to the logical product of the complements:

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$

● **Second Theorem (NAND Gate Equivalence):** The complement of a logical product is equal to the logical sum of the complements:

$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

#### Universal Logic Gates:

**NAND Gate and NOR Gate** are called **Universal Gates** because any conceivable Boolean function (AND, OR, NOT, XOR, XNOR) can be implemented using NAND gates alone or NOR gates alone.

### NAND Implementation of Fundamental Gates:

1.  $\text{NOT}(A) = \overline{A \cdot A}$  (1 NAND gate)
  2.  $\text{AND}(A, B) = \overline{\overline{A \cdot B}}$  (2 NAND gates)
  3.  $\text{OR}(A, B) = \overline{\overline{A} \cdot \overline{B}}$  (3 NAND gates)
  4.  $\text{NOR}(A, B) = \overline{\overline{A} + \overline{B}}$  (4 NAND gates)
  5.  $\text{XOR}(A, B) = A\bar{B} + \bar{A}B$  (4 NAND gates)
- 

### Problem:

Simplify the Boolean expression  $Y = AB + A\bar{B} + \bar{A}B$ .

---

### Step 1 🚀 : Factor out common terms

$$Y = A(B + \bar{B}) + \bar{A}B$$

### Step 2 ⚡ : Apply Complement Law ( $B + \bar{B} = 1$ )

$$Y = A(1) + \bar{A}B = A + \bar{A}B$$

### Step 3 🎯 : Apply Distributive Law ( $A + \bar{A}B = A + B$ )

$$Y = (A + \bar{A})(A + B) = (1)(A + B) = A + B$$

🎯 Simplified:  $Y = A + B$  (Reduces from 3 AND gates to a single OR gate!).

### University Question (6 Marks):

Verify De Morgan's Second Law:  $\overline{A \cdot B} = \bar{A} + \bar{B}$  using a complete truth table.

### Step 1 🚀 : Truth Table Construction

| A | B | $A \cdot B$ | $\overline{A \cdot B}$ (NAND) | $\bar{A}$ | $\bar{B}$ | $\bar{A} + \bar{B}$ (Bubbled OR) |
|---|---|-------------|-------------------------------|-----------|-----------|----------------------------------|
| 0 | 0 | 0           | 1                             | 1         | 1         | 1                                |
| 0 | 1 | 0           | 1                             | 1         | 0         | 1                                |
| 1 | 0 | 0           | 1                             | 0         | 1         | 1                                |
| 1 | 1 | 1           | 0                             | 0         | 0         | 0                                |

### Step 2 ⚡ : Conclusion

Columns 4 and 7 are identically matching for all 4 possible input combinations!

🎯 Verified:  $\overline{A \cdot B} \equiv \bar{A} + \bar{B}$  (NAND gate is electrically equivalent to Bubbled OR).

### University Question (10 Marks):

Show that a 2-input XOR gate ( $Y = A \bar{B} + \bar{A} B$ ) can be synthesized using exactly four 2-input NAND gates with zero redundant gates. Provide algebraic proof for each gate output.

### Step 1 🚀 : Gate 1 (Central Cross NAND)

$$G_1 = \overline{AB}$$

### Step 2 ⚡ : Gates 2 and 3

Gate 2 inputs are A and  $G_1$ :

$$G_2 = \overline{A \cdot G_1} = \overline{A \cdot \overline{AB}} = \bar{A} + \overline{\overline{AB}} = \bar{A} + AB =$$

$$(\bar{A} + A)(\bar{A} + B) = \bar{A}(A + B)$$

Gate 3 inputs are B and  $G_1$ :

$$G_3 = \overline{B \cdot G_1} = \overline{B \cdot \overline{AB}} = \bar{B} + AB = (\bar{B} + B)(\bar{B} + A) = A + \bar{B}$$

### Step 3 🌀: Gate 4 (Output Gate combining $G_2$ and $G_3$ )

$$Y = \overline{G_2 \cdot G_3} = \overline{(\bar{A} + B)(A + \bar{B})}$$

Expand the inner product:

$$(\bar{A} + B)(A + \bar{B}) = \bar{A}A + \bar{A}\bar{B} + BA + B\bar{B} = 0 + \bar{A}\bar{B} + AB + 0 = AB + \bar{A}\bar{B}$$

Apply De Morgan's on the outer inversion:

$$Y = \overline{AB + \bar{A}\bar{B}} = \overline{AB} \cdot \overline{\bar{A}\bar{B}} = (\bar{A} + \bar{B})(A + B) = \bar{A}B + A\bar{B} = A \oplus B$$



Proved: Minimum 4 NAND gates synthesize  $Y = A \oplus B = A\bar{B} + \bar{A}B$ .

## University Solved Examination Questions (Unit 4)

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### MMDU Official University Solved Examination Questions (Unit 4)

**Course:** Basic Electrical & Electronics Engineering ('BELE-001')

**Pattern:** Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (6 Marks In-Depth), D (6 Marks Design/Long)

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**EASY**

### Question 10 (1 Mark): Self-Complementing Property of Excess-3 Code

#### Question:

Demonstrate why Excess-3 code is termed a self-complementing code using decimal digit 4.

#### Question Schematic / State Specification:

...

Question 9's Complement Inversion: 9's complement of decimal 4 is  $(9 - 4) = 5$ .

...

#### Step-by-Step Solution:

#### Solution Analytical Derivation & Schematic:

...

Solution Bitwise Inversion Trace:

`Excess-3(4) =  $4 + 3 = 7 = 0111_2$ `.

`Bitwise NOT(0111\_2) =  $1000_2 = 8_{10} = 5 + 3 = \text{Excess-3}(5)$ `!

...

Taking the 1's complement of an Excess-3 codeword directly produces the Excess-3 representation of its 9's complement, greatly simplifying digital decimal subtraction arithmetic.



**Verified Result:** Question 10 step-by-step resolution per MMDU university marking scheme.

EASY

### Question 11 (1 Mark): Why NAND is Termed a Universal Gate

#### Question:

Prove that a NOT gate can be synthesized using a single 2-input NAND gate.

#### Question Schematic / State Specification:

...

Question Input Tie-Together: Connect both inputs of NAND together to input signal A.

...

#### Step-by-Step Solution:

#### Solution Analytical Derivation & Schematic:

...

Solution Boolean Idempotence Formulation:

$Y = (A \cdot A)' = A'$  (By Boolean Idempotent Law).

...

By connecting both inputs of a 2-input NAND gate to common signal A, the output becomes  $(A \cdot A)' = A'$ , functioning identically to an inverter (NOT gate).



**Verified Result:** Question 11 step-by-step resolution per MMDU university marking scheme.

EASY

### Question 12 (1 Mark): Gray Code Usage in Rotary Optical Encoders

#### Question:

Why is Gray code preferred over standard Binary code in optical shaft position encoders?

#### Question Schematic / State Specification:

...

Question Binary Transition Glitch: Binary 7 (0111) → 8 (1000) flips 4 bits simultaneously; Gray flips only 1 bit.

...

#### Step-by-Step Solution:

#### Solution Analytical Derivation & Schematic:

...

Solution Unit Distance Property:

Hamming Distance = 1 between adjacent counts ⇒ eliminates intermediate spurious read codes.

...

In standard binary, multiple bits switch simultaneously (e.g. 0111 to 1000). Due to physical sensor misalignment, bits switch at slightly different nanoseconds, creating erroneous intermediate spikes. Gray code changes only one bit at a time, preventing read errors.



**Verified Result:** Question 12 step-by-step resolution per MMDU university marking scheme.

**MEDIUM**

### Question 16 (2 Marks): 2's Complement Subtraction

#### Question:

Perform subtraction  $(28)_{(10)} - (15)_{(10)}$  using 8-bit 2's complement arithmetic.

#### Question Schematic / State Specification:

...

Question Binary Operand Conversion:  $28 = 00011100_2$  |  $15 = 00001111_2$ .

...

#### Step-by-Step Solution:

#### Solution Analytical Derivation & Schematic:

...

Solution 2's Complement Addition Trace:

`1's Complement of 15:  $11110000_2$ `

`2's Complement of 15:  $11110000 + 1 = 11110001_2$ `

`Add to 28:  $00011100 + 11110001 = 1\ 00001101_2$ `

`Discard End Carry (1)  $\Rightarrow$  Result =  $00001101_2 = 8 + 4 + 1 = (13)_{(10)}$  (Correct!)`

...

The end carry indicates a positive result, yielding  $00001101_2 = 13_{10}$ , matching  $28 - 15 = 13$ .



**Verified Result:** Question 16 step-by-step resolution per MMDU university marking scheme.

**HARD**

### Question 20 (4 Marks): 4-Variable K-Map Minimization with Don't Cares

#### Question:

Minimize the Boolean function:  $F(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + d(0, 2, 5)$  using a 4-variable K-Map. Implement using basic gates.



#### Question Schematic / State Specification:

...

Question K-Map Cell Mapping: Minterms at 1, 3, 7, 11, 15; Don't cares at 0, 2, 5.

...

#### Step-by-Step Solution:



#### Solution Analytical Derivation & Schematic:

...

Solution Optimal Grouping Strategy:

`Group 1 (Octet): Combine minterms  $m_1, m_3, m_7, m_{11}, m_{15}$  with don't cares  $d_0, d_2, d_5$ !

`Cells (0, 1, 2, 3, 5, 7, 11, 15): Notice column CD = ( $m_3, m_7, m_{11}, m_{15}$ ) forms a Quad CD`

`Row A'B' (0, 1, 2, 3) + (5, 7): Combine (1, 3, 5, 7) with (3, 7, 11, 15)  $\Rightarrow$ `

`Quad 1: (3, 7, 11, 15) yields CD`

`Quad 2: (1, 3, 5, 7) yields A'D`

`Quad 3: (0, 1, 2, 3) with  $d_0, d_2$  yields A'B`

`Minimal SOP Expression:  $F = CD + A'B' + A'D$  (or  $F = CD + A'B'$ )`

...

By leveraging Don't Care cells  $d(0, 2)$ , the function simplifies down to  $F = CD + A'B'$ , eliminating 3 gates.



**Verified Result:** Question 20 step-by-step resolution per MMDU university marking scheme.

